

BIOLOGY GRADE 11: CELL DIVISION

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.4 (10 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	11
Subject	Biology
Strand	Strand 1.0: Cell Biology and Biodiversity
Sub-Strand	Sub-Strand 1.4: Cell Division
Total Duration	10 lessons × 40 minutes = 400 minutes total
Sub-Strand Content	– Meaning of cell division and structure of chromosomes in organisms – Types of cell division (mitosis and meiosis) – Significance of cell division in organisms (include tissue culture) – Budding and binary fission
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) explain the significance of mitosis and meiosis in organisms b) illustrate the stages of mitosis and meiosis in cells c) distinguish between budding and binary fission in relation to cell division d) recognise the importance of cell division in reproduction, growth, and development.
Core Competencies	– Creativity and Imagination: The learner innovatively models budding in yeast and binary fission in bacteria. – Digital Literacy: The learner enhances computing skills while watching animations showing various stages of mitosis and meiosis. – Communication and Collaboration: Learners work in groups to observe cell diagrams, build clay models and present findings to peers. – Critical Thinking and Problem Solving: Learners analyse chromosome numbers and case studies to decide whether mitosis or meiosis is involved. – Learning to Learn: Learners revise their initial ideas and models as new evidence emerges, and track their progress on the Driving Question Board. – Self-efficacy: Learners build confidence by explaining their own growth using evidence they gathered themselves.
Core Values	– Unity: The learner collaborates with others while using a light microscope to observe stages of meiosis and mitosis in living tissues. – Respect: The learner appreciates others' efforts while presenting illustrations to show the behaviour of chromosomes in various stages of cell division. – Integrity: Learners record observations honestly, even when these differ from expected results or from the textbook.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners generate questions about their own growth and post them on the Driving Question Board.

	– Developing and Using Models: Learners build clay models of 1 cell → 2 cells, draw chromosome behaviour and revise a cumulative model each lesson. – Planning and Carrying Out Investigations: Learners prepare and observe onion root tip slides under a light microscope. – Analyzing and Interpreting Data: Learners analyse cell counts, observation tables and chromosome numbers before and after division. – Using Mathematics and Computational Thinking: Learners calculate chromosome numbers after fertilisation ($46 + 46 = 92$ versus $23 + 23 = 46$). – Constructing Explanations: Learners explain how mitosis and meiosis account for growth, repair and reproduction. – Engaging in Argument from Evidence: Learners justify with evidence which type of cell division operates in each case study. – Obtaining, Evaluating and Communicating Information: Learners research chromosome structure and stage names from texts and animations, then present to peers.
Pertinent & Contemporary Issues (PCIs)	– Safety and Security: The learner observes safety precautions while using a light microscope to observe stages of meiosis and mitosis in living tissues. – Communicable Diseases: The learner recognizes the importance of healthy practices for prevention of communicable diseases while modelling budding in yeast and binary fission in bacteria.
Career Connections	– Medical Doctor / Paediatrician: monitors child growth and development, which depends on cell division. – Oncologist / Cancer Researcher: studies uncontrolled mitosis in cancer, e.g. at Kenyatta National Hospital. – Geneticist / Genetic Counsellor: applies knowledge of meiosis to inheritance and chromosome disorders. – Plant Tissue Culture Technician: multiplies disease-free bananas, tea and potatoes (e.g. KALRO, Kericho tea estates). – Laboratory Technologist / Microbiologist: studies bacterial binary fission and yeast budding. – Agricultural Officer / Livestock Breeder: uses reproduction and cell division knowledge to improve crops and herds.
Focus for Lessons	This unit uses the puzzle of a child growing from infancy to adolescence to drive learners to discover that cells divide to produce new cells. Learners gather evidence from diagrams, models, readings, animations and microscope work to build understanding of chromosomes, mitosis, meiosis and asexual reproduction (budding and binary fission). Names of stages and terms are introduced only after learners have described what they observe, and learners revise a class model across the unit.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How does the body produce the enormous number of new cells needed for growth? KICD KEY INQUIRY QUESTIONS: 1. How is cell division important?
Anchoring Phenomenon	A child grows from infancy to adolescence. Learners compare baby photographs of themselves with how they look now and watch a time-lapse of a child growing up over 14 years. A newborn baby weighing about 3 kg becomes a teenager of 50 kg or more, so the body must produce an enormous number of new cells. It is puzzling because we only eat food, yet our bodies make billions of new cells. Where do all these cells come from, and how does each new cell get the right instructions to work properly?
Storyline Thread	Lesson 1: Students are introduced to the phenomenon. They watch a time-lapse of growing up over 14 years, make initial drawings and write their initial ideas on fullscaps, then generate questions and post them on the Driving Question Board (How does growth occur?). Lesson 2: Meaning of cell division and structure of the chromosome. Groups observe and analyse diagrams of cells before and after division and record counts in a table. They build a 1 cell → 2 cells clay model and propose what must be passed on, then label an unlabelled chromosome diagram using a reading. Question raised: What happens to the chromosomes during cell division to ensure that each new cell gets the correct genetic information?

	Lesson 3: Types of cell division: mitosis and its stages. Groups describe chromosome movement in a mitosis diagram first, then match their observations to prophase, metaphase, anaphase, telophase and cytokinesis using a reading. They return to the phenomenon to explain how mitosis helps a child grow. Question raised: If mitosis produces new body cells similar to the original, how are the cells used for sexual reproduction produced? Lesson 4: Meiosis and its stages. Groups solve the chromosome-number problem ($46 + 46$), sequence unlabelled pictures of meiosis I and II, use a reading and videos to label the stages, then explain why sperm and egg cells have 23 chromosomes. Question raised: What would happen to an organism if mitosis and meiosis did not occur? Lesson 5: Significance of mitosis and meiosis. Groups investigate case studies (child growing, skin healing, plant roots and shoots, chicken producing offspring, parents producing children with different characteristics) and complete a table linking each to mitosis or meiosis with evidence. Question raised: How can a single organism produce a new organism without producing sex cells? Lesson 6: Budding and binary fission. Learners compare the two processes, model budding in yeast and binary fission in bacteria, present the main differences, and conclude that both are asexual reproduction from a single parent without fusion of sex cells. Lesson 7: Microscope practical on mitosis. Learners prepare and observe onion root tip slides with a light microscope following safety precautions, identify stages of mitosis, and link their observations to growth in the phenomenon. Lesson 8: Significance of cell division in organisms, including tissue culture. Learners explore how cell division supports growth, repair, reproduction and development, and how tissue culture multiplies plants such as bananas and tea in Kenya. Lesson 9: Consolidation and digital animations. Learners watch animations of mitosis and meiosis, compare mitosis and meiosis in a table, and present their chromosome-behaviour models and illustrations to peers, giving respectful feedback. Lesson 10: Final model and return to the phenomenon. Learners complete and present their final cumulative model, revisit the Driving Question Board and their initial drawings, and write an evidence-based explanation of how the body produces the enormous number of new cells needed for a child to grow.
Supporting Phenomena	– A cut on the skin of a child playing football in Kisumu heals within days, with new skin cells replacing the damaged ones. – A maize seedling or sukuma wiki plant produces new roots and shoots, and a farmer in the Rift Valley sees crops grow taller each week. – Brothers and sisters from the same parents look similar but are not identical, and sperm and egg cells must each carry 23 chromosomes. – Yeast used to make mandazi dough rises and bubbles as it multiplies, and a single bacterium can become millions in hours, so one organism can make new organisms without sex cells.

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LESSON 1 (80 minutes (double lesson)): From Newborn to Teenager: Where Do All the New Cells Come From?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the unit by confronting learners with the puzzle of how a 3 kg newborn becomes a 50 kg teenager, so that learners record their initial ideas and drawings and raise questions that will drive the whole Cell Division unit.
Knowledge	Learners describe that growth from infancy to adolescence involves a very large increase in the number of cells. They begin to explain the meaning and importance of cell division (KICD Key Inquiry: How is cell division important?). Learners recognise that they do not yet know where the new cells come from or how each new cell gets the right instructions.
Skills	Developing and using models (initial drawn model); asking questions; planning and carrying out observation of a time-lapse; communicating ideas orally and in writing; collaborating in groups. Digital Literacy: watching and interpreting a time-lapse video.
Attitudes	Unity: learners collaborate with peers when sharing and grouping questions. Respect: learners listen to and appreciate others' ideas and drawings during the gallery walk. Curiosity and willingness to say 'I do not know yet'.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lesson 1 introduces the anchoring phenomenon and opens the Driving Question Board (DQB) with the question 'How does growth occur?'. Learners' initial models and questions are the baseline revisited in every later lesson (mitosis, meiosis, chromosomes, budding, binary fission, tissue culture). The next lesson begins by investigating: How does growth occur?
Safety Notes	Class is large, so keep aisles clear when learners move for the gallery walk and when posting on the DQB. Move in waves of 10 learners. Use only pencils and paper. If baby photographs are brought, handle them gently and respect learners who do not have or do not wish to share photos (Safety and Security; Respect). If using a projector or phone, keep cables off the floor.

B. LESSON OVERVIEW

Learners compare baby photographs of themselves (or recall their birth stories from home, e.g. a mother in Kisumu or Nairobi saying 'you were only 3 kg') with how they look now, then watch a time-lapse of a child growing over 14 years. They predict what happens inside the body for a 3 kg newborn to become a 50 kg teenager, draw an initial model on fullscap, and write their initial ideas. Using Think-Pair-Share, a Discussion Diamond, a Gallery Walk and a Jigsaw for question sorting, they generate questions and post them on the DQB under 'How does growth occur?'. Because the class is large, work is done in groups of 5 to 6, with shared roles and with posting done by group representatives. The lesson ends with an exit ticket and leads into Lesson 2.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Comparing mitosis and meiosis	Learners take out their baby photograph or recall a family story of their birth weight (about 3 kg). They compare it with how they look now (about 50 kg or more). In	Say: 'Today we begin a mystery about your own bodies. Take out your baby photo, or think of a baby in your family.' Display on the board: 'A newborn weighs about 3	Think-Pair-Share and Predict-Observe-Explain (Predict step). Silent individual thinking gives every learner, even in a large class, a chance to commit to an	Oral questions: listen for whether learners mention cells, food, size, or number. Observable indicators: (1) learner writes a prediction with at least one reason; (2) learner

Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Cancerous Cell Growth (PLIX) Source: CK-12 Search ARES for similar readings	pairs they discuss: 'You only eat ugali, sukuma wiki, githeri and beans. How does a 3 kg baby become a 50 kg teenager?' Then on a fullscap sheet they individually write a prediction: what is happening inside the body for this growth to occur? Learners who have no photograph use the memory of a younger sibling or cousin.	kg. You now weigh about 50 kg. Where did the extra 47 kg come from?' Give WAIT TIME of 10-15 seconds of silent thinking before any talking. Say: 'Turn to your partner and share for 2 minutes.' Circulate and listen for ideas like 'cells get bigger', 'food turns into the body', 'cells multiply'. Cold-call 4 pairs: 'What did your partner say that you had not thought of?' Do not correct any answers; say 'Thank you, we will test that idea.' Class is large, so pairs share with the neighbour in the same row and you sample from different parts of the room. Set the norm: 'There are no wrong ideas today, only starting ideas.'	idea before hearing others. Pair talk lets learners rehearse their reasoning. The recorded prediction gives a baseline to compare with at the end of the unit.	uses the word 'cells' unprompted; (3) learner distinguishes between cells getting bigger and more cells being made. Tally on a class list how many pairs mention 'more cells' versus 'bigger cells'.
Observe Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Growth Mindset: reading realistic fiction: A New Friend 4 Source: Khan Academy (English - US curriculum) Search ARES for similar readings	Learners watch the time-lapse of a child growing up over 14 years (played twice). First viewing: watch silently. Second viewing: in groups of 5-6, record on fullscap what changes they notice (height, mass, face, limbs, hair, teeth) and the body parts involved. Each group member takes a role (observer of head and face, observer of limbs, observer of body size, recorder, reporter). Learners then link what they see to the idea that a 3 kg newborn must produce billions of new cells, and note what puzzles them, for example: 'We only eat food, so where do the new cells come from?' and 'How does every new cell know what to do?'	Say: 'Watch carefully. This child is like any of you growing up in Eldoret, Mombasa, or Kakamega.' Play the time-lapse once without comment. Then say: 'Watch again and record only what you SEE, not what you think.' Pause the video at 3 points (infancy, about 7 years, adolescence) and ask: 'What has changed?' Allow WAIT TIME of 10-15 seconds, then cold-call 3 learners at each pause. Share the key fact: 'The body of a newborn is about 3 kg; by about 14 years it may be 50 kg or more. Your body is made of trillions of cells.' Ask: 'Is the baby just a small copy of an adult with the same cells getting bigger, or are there more cells?' Take 3 responses without judging. Circulate to each group, asking: 'What evidence in the video supports your idea?' If there is no electricity or projector, narrate with a set of 4 sequenced photographs or a chart drawn on manila paper, and read aloud descriptive notes about the child's growth.	Predict-Observe-Explain (Observe step) and Jigsaw roles within groups. Assigning each learner a specific feature to observe ensures all learners in a large class contribute distinct evidence. The group then pieces together the individual observations into a whole picture of growth, mirroring a jigsaw.	Observable indicators: recorders list at least 4 specific changes; groups separate observation ('he becomes taller') from inference ('cells are multiplying'); learners raise a question not answered by the video. Use a quick model check: walk around and note which groups mention 'number of cells' or 'many cells'. Oral question: 'What in the video is the best evidence that the body is producing new material?'

Explain Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Linear vs. exponential growth: from data Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Individually, learners draw their initial model on fullscap: a labelled drawing showing a baby and a teenager, and what they think is happening inside the body (for example cells, food entering the body, new cells appearing). They write a short caption of 2-3 sentences explaining how a 3 kg baby becomes a 50 kg teenager. Next, groups form a Discussion Diamond: each group ranks 4 or 5 statements from most to least convincing, for example (a) 'Cells grow bigger only', (b) 'The body makes new cells from food', (c) 'New cells come from old cells', (d) 'Each baby is born with all the cells it will ever need'. Groups then do a Gallery Walk to view other groups' drawings and diamonds, leaving one sticky note or pencil comment of 'I notice...' or 'I wonder...'.	Say: 'Draw what you think, not what you think I want. Show what is happening inside the body. Use arrows and labels.' Give 8 minutes of silent drawing. Display the Diamond statements on the board or on a printed strip. Say: 'In your group, agree on the order, with the most convincing at the top. You must give a reason from the video or from your life, for example how a child eating ugali and sukuma wiki daily grows.' Give WAIT TIME of 10-15 seconds after each prompt before taking contributions. For the Gallery Walk: put drawings on desks or walls, split the class into 2 halves so only half walks at a time, and give 6 minutes per half to keep aisles clear. Rule: 'Be respectful: comment on ideas, not on the person.' After the walk, cold-call 5 learners: 'What similarity did you see across drawings? What was different?' Record patterns on the board (e.g. 'many drawings show cells getting bigger; few show cells dividing'). Do NOT give the answer; say 'These are our starting models; we will test them.'	Discussion Diamond and Gallery Walk. The Diamond forces learners to weigh competing explanations and justify rankings with evidence, surfacing misconceptions such as 'cells only grow bigger'. The Gallery Walk lets all learners compare models across groups without every group presenting aloud, which saves time in a large class and builds Respect and Unity.	Model checks: look at each initial drawing for (1) cells shown, (2) any mechanism for increasing cell number, (3) the role of food, (4) labels and arrows. Record common misconceptions (cells only enlarge; food directly becomes body parts; babies born with all cells). Oral questions during the Diamond: 'Which statement did you place at the top and what evidence supports it?' Observable indicators: learners give reasons using 'because' and cite the video or personal experience.
Driving Question Board (DQB) Creation  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Exponential vs. linear growth over	Each learner writes 2 questions on separate strips of paper about what puzzled them, for example: 'How do cells make more cells?', 'Why do we stop growing at about 20 years?', 'How does a cell know what to become, bone or skin?', 'Why do wounds heal?', 'How do twins form?', 'How does each new cell get the right instructions?' Groups pool their questions, sort them into categories (how new cells are made, instructions in cells, why growth stops, healing, reproduction), choose their 3 best, and the group reporter posts them	Before the lesson, prepare a large board (manila paper on the wall) titled 'DRIVING QUESTION: How does the body produce the enormous number of new cells needed for growth?' with the sub-heading 'How does growth occur?'. Say: 'Write questions that start with How, Why or What. Questions we can investigate.' Model one: 'How does a single cell become billions?' Give WAIT TIME of 10-15 seconds, then 3 minutes for individual writing. Because the class is large, only group reporters come forward in 3 waves of about	Jigsaw (question sorting) with the DQB. Each group becomes an expert on one category of questions, and the combined board shows the whole class's thinking. This shares the workload in a large class and gives learners ownership of the learning path.	Observable indicators: questions are testable (How/Why), connect to the phenomenon, and go beyond yes/no; groups sort questions into logical categories; at least one question per group mentions cells multiplying, instructions, or why growth stops. Oral questions: 'How does your question connect to the child in the video?' Count the questions per category to guide Lesson 2 planning.

time Source: Khan Academy (English - US curriculum) Search ARES for similar readings	on the Driving Question Board under the title 'How does growth occur?'. A Jigsaw is used to sort: each group takes one category and checks the board for duplicates.	8 groups to post. Cold-call 4 groups to read one question aloud and explain why it matters. Group similar questions, ask: 'Which of these could we answer by looking through a microscope? Which need a model?' Mark the questions linked to the KICD inquiry 'How is cell division important?'. Say: 'We will return to this board in every lesson and move questions to answered as we gather evidence.' Highlight the question for next lesson: 'How does growth occur?'		
Model Building Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Cancerous Cell Growth (PLIX) Source: CK-12 Search ARES for similar readings	Learners return to their initial drawing and add a section titled 'What I am not sure about', listing 2 things they cannot yet explain. They then write on a fullscap exit ticket: (1) one idea from today that you still believe, (2) one idea you are now unsure of, (3) one question you most want answered. Learners keep their fullscap in their portfolio to be revised in later lessons. They also start a Summary Table with the columns: What we observed / What we learned / How we can explain it.	Say: 'This is Version 1 of your model. Scientists like Prof. Wangari Maathai, who trained as a biologist in Nairobi, began with questions and kept improving their ideas. We will revise this model many times.' Give WAIT TIME of 10-15 seconds before learners write their exit tickets. Collect exit tickets at the door, 1 row at a time, to avoid crowding. Cold-call 3 learners to share the gap they spotted in their model. Tell the class: 'Next lesson we ask: How does growth occur? We will look for evidence of how the body makes new cells.' Store the initial models in labelled folders or envelopes by group. After class, sort exit tickets into 'cells bigger', 'more cells', and 'unsure' to plan Lesson 2.	Predict-Observe-Explain (Explain, initial model) with Exit Ticket reflection. Naming the gaps in their own model makes learners aware of what they need to learn, which increases motivation and connects directly to the DQB.	Exit tickets: look for (1) a clear statement of an initial belief, (2) an honest identification of uncertainty, (3) a question linked to cell number or instructions. Model check: initial drawings are dated, labelled and stored. Success indicator: at least 80% of exit tickets list a specific unanswered question about how new cells are made.

D. TEACHER REFLECTION

Which misconceptions were most common in the drawings (cells only grow bigger, food directly becomes body parts)? Did the large class size limit participation, and which groupings (rows, pairs, groups of 5-6) worked best? Did learners without baby photographs feel included? Which DQB questions are testable in the next lessons using available resources (charts, models, microscope slides of onion root tip, yeast)? How will I use the exit tickets to begin Lesson 2 with the question 'How does growth occur?'

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We watched a time-lapse of a child growing over 14 years and compared baby photographs with how we look now. A newborn of about 3 kg became a teenager of 50 kg or more.
What did I learn?	Growth involves a huge increase in the number of cells, and eating food alone does not explain where the new cells come from. We do not yet know how new cells are made or how each gets the right instructions.
How does this explain the phenomenon?	Our first ideas are that cells might grow bigger, multiply, or be made from food. We posted questions on the DQB under 'How does growth occur?' and will test our initial models in the next lessons.

LESSON 2 (80 minutes (double lesson)): What Is Cell Division? Meaning of Cell Division and the Structure of the Chromosome

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence that growth requires new cells made by existing cells dividing, and identify the chromosome as the structure carrying instructions that must be passed on to new cells.
Knowledge	By the end of the lesson, the learner should be able to: explain the meaning of cell division (one parent cell producing new daughter cells); illustrate and label the structure of a chromosome (chromatids, centromere, DNA/genes); and state that chromosomes carry the genetic information each new cell needs to function.
Skills	Observing and counting from diagrams; recording data in a table (SEP: Analyzing and Interpreting Data); building and using a physical model (SEP: Developing and Using Models); constructing explanations from evidence (SEP: Constructing Explanations); researching from a reading and communicating findings; collaborating in groups.
Attitudes	Unity: the learner collaborates with others in group tasks. Respect: the learner appreciates others' efforts while presenting models and diagrams. Curiosity about how a 3 kg newborn becomes a 50 kg teenager.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lesson 1 established that a child's body grows from a 3 kg newborn to a teenager of 50 kg or more, so billions of new cells are needed. Lesson 2 provides the first mechanism: existing cells divide to make new cells, and chromosomes carry the instructions. This raises the question for Lesson 3: What happens to the chromosomes during cell division to ensure that each new cell gets the correct genetic information?
Safety Notes	Clay and diagrams only, so risks are low. Remind learners to keep clay off the floor (slip hazard) and to wash hands after handling it. If using phones or computers for research, follow school rules on safe internet use. Keep walkways clear during the gallery walk. For large classes, assign group roles to avoid crowding.

B. LESSON OVERVIEW

Learners return to the phenomenon of a child growing from infancy to adolescence (their own baby photographs versus now). They predict where new cells come from, then analyse diagrams of cells before and after division and record counts in a table. In groups of 5-6 (suited to a large class, with shared materials) they build a 1 cell → 2 cells clay model and propose what must be passed on. Using a reading (the resource supplied by the teacher), a jigsaw approach has each group member become an expert on one chromosome part, and groups label an unlabelled chromosome diagram and write the function of the main structures. A gallery walk and discussion diamond let learners compare models and ideas. They add questions to the Driving Question Board and revise the class model, ending with an exit ticket and the new question: What happens to the chromosomes during cell division to ensure that each new cell gets the correct genetic information?

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners hold up a baby photograph (or recall one) and	Say: "Last lesson we saw that a newborn of about 3 kg becomes a	Think-Pair-Share and Predict-Observe-Explain (Predict step):	Oral questions and observation: note how many learners draw (a)

 VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: cell division parent daughter	compare it with how they look now. Individually, in 3 minutes, they write on a sticky note or paper: 'A baby of 3 kg becomes a teenager of 50 kg. Where do the extra billions of cells come from? Draw what you think happens to ONE cell when the body makes more cells.' They then think-pair-share and keep their predictions for later comparison.	teenager of 50 kg or more. We only eat ugali, sukuma wiki and beans. So where do billions of new cells come from?" Give each learner time to write, then WAIT TIME of 10-15 seconds in silence before pair talk. Say: "Turn to your partner. Share your drawing and listen to theirs for 2 minutes." Cold-call 4 pairs from different parts of the room: "What did your partner say? What did you agree or disagree on?" Record 4-5 prediction types on the board without saying which is correct, e.g. 'cells come from food', 'cells grow bigger only', 'cells split'. Say: "Today we collect evidence to test these ideas." For a large class, only collect predictions from front, middle and back rows and have groups keep their sticky notes.	writing individually first ensures every learner commits to an idea; pair talk rehearses reasoning; the recorded predictions give the class something to test with evidence, making the later observations purposeful.	one cell becoming two, (b) a cell only growing larger, (c) food turning into cells. Look for use of words like 'divide' or 'split'. Record the approximate number of learners in each category to guide later questioning.
Observe Phase  VIDEO: Chromosomes, chromatids and chromatin Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: No closely matching reading in the ARES library for this activity.  Search ARES for readings Search terms: cell division parent daughter	Activity 1 (20 min): In groups of 5-6, learners receive a set of printed diagrams of cells at different stages, including a cell before and after division. They observe carefully, identify similarities and differences, count the cells before and after division, describe what appears to happen to the original cell, and record observations in a table (columns: Diagram, Number of cells before, Number of cells after, What I notice). Evidence: an observation table showing that one cell can produce new cells through division. Activity 2 (20 min): Groups use modelling clay to build 1 cell → division → 2 cells. Each group discusses and writes: 'What must be passed from the original cell to the new cells so they can function correctly?' They propose their own explanations BEFORE any chromosome teaching, and add the idea to the model (e.g. a	Distribute one diagram set per group; for a large class, give one set to every 5-6 learners and use A3 pages or projected copies on the board. Assign roles: materials manager, recorder, counter, reporter, timekeeper, and a questioner. Say: "Look at the first diagram. How many cells do you see before? How many after? Write it down before you talk." Circulate and ask: "What happened to the original cell? Where did it go?" and "How could one cell make two cells that both look the same?" Use WAIT TIME of 10-15 seconds after each question. For the clay task say: "Make one round cell with a nucleus. Now show me what happens so that you end up with two cells. What must the new cells receive so they can work like the first one, the way a child of Kisumu or Nairobi grows up healthy with all	Predict-Observe-Explain (Observe step) with Discussion Diamond: after counting, each learner ranks the four statements 'cells only get bigger', 'cells divide to make new cells', 'cells appear from food', 'cells are replaced one by one from outside' in a diamond (most to least supported by the evidence) individually, then group negotiates a shared diamond. This builds evidence-based reasoning and shows that data, not opinions, decide ideas.	Model checks: look for (1) a clay model showing one cell becoming two, (2) nucleus or a hereditary material shown in both new cells, (3) a written explanation that mentions 'instructions' or 'information' passed on. Observation table check: counts correct (1 → 2, and 2 → 4 if shown) and description uses 'divides' or 'splits'. Use a quick thumbs check: 'Does your model show both new cells with something inside?'

	clay 'nucleus' with colored thread or beads inside each cell). Evidence: a learner-generated model and explanation.	parts working?" Do NOT mention chromosomes or DNA yet. Cold-call 3 groups to show their models and say: "Tell us what you think is passed on." Record all ideas on the board (e.g. 'the nucleus', 'instructions', 'something inside the nucleus').		
Explain Phase  VIDEO: Chromosomes, chromatids and chromatin - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: cell division parent daughter	Activity 3 (25 min): Jigsaw. Each home group of 6 receives a large unlabelled chromosome diagram. Learners first observe, identify visible structures and discuss what each part might represent. Then they split into expert groups, each studying one section of the reading (the resource provided): Expert A - the chromatid, Expert B - the centromere, Expert C - DNA and genes, Expert D - chromosome numbers (humans have 46 chromosomes; maize has 20), Expert E - the role of chromosomes in passing information during division, Expert F - the arms and sister chromatids. Experts return to home groups and teach their part. Groups label the diagram, write the function of the main structures, and connect it to their clay models: 'What did our model not show?' Evidence: a labelled chromosome diagram and short explanations of its parts. They then complete the Predict-Observe-Explain explanation: 'How does one cell become many for a growing child?' Gallery walk (10 min): groups display models and diagrams and rotate, leaving one comment (a star and a question) on each.	Say: "Before reading, what do you think each part of this diagram does? Talk for 2 minutes." Use WAIT TIME of 10-15 seconds before taking ideas; cold-call 3 groups. Then give the reading and say: "Read your section twice. Underline the function. You have 8 minutes to become an expert." Move between expert groups asking: "How does this part help a new cell get the right instructions?" Misconceptions to watch for: chromosomes are separate from DNA; the centromere is a type of cell; chromatids are two different chromosomes. Correct by pointing to the diagram: "Look at the two identical arms joined at one point." Say for the gallery walk: "Leave a star for a strength and a question for something you wonder. Walk clockwise, 2 minutes per display." For a large class, run two gallery loops with half of the groups standing near their display while others rotate. Close with 3 cold-calls: "Who can explain the meaning of cell division in your own words, using one example from a growing child?" Confirm: cell division is the process by which a parent cell divides to form new daughter cells; chromosomes carry the genetic information.	Jigsaw + Gallery Walk: jigsaw gives every learner responsibility for knowledge that others need, which builds interdependence (Unity) and gets reading done at speed in a large class; the gallery walk lets learners compare their labelled diagrams and models with others', practise respectful feedback (Respect), and sharpen the idea that chromosomes carry the instructions passed to new cells.	Oral questions and diagram check: ask 'Where is the centromere? What does it join?' and 'What does DNA carry?' Observable indicators: correct labels (chromatid, centromere, DNA/gene regions), a sentence linking chromosomes to genetic information, and the star/question comments on the gallery sheets. Exit ticket (last 5 min, individual): 1) Define cell division in one sentence. 2) Draw and label a chromosome with at least three parts. 3) Write one question about what happens to chromosomes when a cell divides.
Driving Question Board (DQB)	Learners revisit the Driving Question: 'How does the body produce the enormous number of	Say: "Look at the board. Which of our earlier questions can we now partly answer? Which new	Discussion Diamond (question prioritisation): groups rank their questions by how much they would	Oral questions and note counting: check that at least 80% of groups submit a question that refers to

Creation  VIDEO: DNA as the "transforming principle" Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: cell division parent daughter	new cells needed for growth?' Each group writes 1-2 new questions on sticky notes from today's evidence (e.g. 'How do chromosomes copy themselves?', 'How does each new cell get the same number of chromosomes?', 'Why do maize and humans have different chromosome numbers?'). They place the notes on the DQB under categories such as 'What is passed on?', 'How is it passed on?', 'Why does it matter?'. They also mark which earlier questions are partly answered (e.g. 'Where do new cells come from?').	questions have appeared?" Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before sharing. Cold-call 4 groups to read one question each and say: "Why is that question important for explaining growth?" Cluster similar questions and highlight the one that links to next lesson: "What happens to the chromosomes during cell division to ensure that each new cell gets the correct genetic information?" Place a star next to it. Mark answered questions with a tick and a date. In a large class, one group representative places the notes while others vote by raising hands.	help explain the phenomenon, then share. This builds student ownership and shows that the next lesson comes from the class's own needs, not only the teacher's plan.	chromosomes or instructions; note questions that show misconceptions (e.g. 'Do cells eat food to divide?') for follow-up. Observable indicator: questions use evidence words such as 'chromosome', 'copy', 'pass on', 'daughter cells'.
Model Building Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  READING: <i>No closely matching reading in the ARES library for this activity.</i>  Search ARES for readings Search terms: cell division parent daughter	Learners begin or revise the class's cumulative model explaining how a child grows. On a poster or notebook page, they draw: baby → child → teenager at the top; below, a cell dividing 1 → 2 → 4 with a nucleus and chromosomes in each; and arrows labelled 'instructions passed on'. They add a labelled chromosome, compare with their initial predictions from the Predict Phase, and note what they changed (e.g. 'Cells do not just come from food; they come from existing cells dividing'). They also write a gap: 'We still do not know how the chromosomes are shared equally.'	Say: "Take out your first prediction. What do you want to keep, change or throw away? Add the new evidence to your model in a different color." Use WAIT TIME of 10-15 seconds, then cold-call 3 learners: "What did you change, and what evidence made you change it?" Ask: "Does your model explain how one Kenyan child of 3 kg becomes 50 kg? What is still missing?" Collect the exit tickets at the door. Conclude: "We know cells divide and chromosomes hold instructions. Next lesson we find out how chromosomes behave so every new cell gets the correct information." Large-class tip: models are drawn individually but one class model is built on the board by 4 volunteers.	Predict-Observe-Explain (Explain/Revise step) and Model Building: learners compare initial predictions with the evidence and revise a visible model, which shows how scientific understanding changes with data and identifies gaps for the next investigation.	Model checks and exit tickets: look for a model showing cell division, nucleus/chromosomes in both daughter cells, a labelled chromosome with at least three parts, an arrow or phrase 'genetic information passed on', and a stated gap. Success indicator: at least 70% of learners correctly define cell division and label chromatid, centromere and DNA on the exit ticket. Use results to group learners for support in Lesson 3.

D. TEACHER REFLECTION

Which groups moved from 'cells just grow bigger' or 'cells come from food' to the idea that existing cells divide, and what evidence convinced them? Did the clay models

show both new cells receiving something, or did some groups only split the cell membrane? Which expert in the jigsaw struggled to explain their part, and why? Was the reading at the right level, and how long did the large class take to move between stations? Which exit ticket questions suggest misconceptions about chromosomes versus DNA? What will I adjust in Lesson 3 so that every learner can explain how chromosomes are passed to daughter cells?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed diagrams where one cell became two (or more) cells after division, and our observation table recorded the counts. We built a clay model of 1 cell → 2 cells and saw that each new cell needs something from the original cell. We observed an unlabelled chromosome diagram with two identical arms joined at one point.
What did I learn?	Cell division is the process by which a parent cell divides to form new daughter cells, which is how the body makes the many new cells needed for growth. Chromosomes carry the genetic information (DNA and genes) that each new cell needs. A chromosome has two chromatids joined at a centromere.
How does this explain the phenomenon?	A child grows from 3 kg to 50 kg because existing cells divide to form new cells, not because food turns directly into cells. For the new cells to work properly, the original cell must pass on chromosomes carrying genetic information. We still need to explain what happens to chromosomes during cell division so that each new cell gets the correct genetic information.

LESSON 3 (80 minutes (double lesson)): Types of Cell Division: Mitosis and Its Stages

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners gather evidence on chromosome movement from a mitosis diagram, then use a short reading to name the stages of mitosis and cytokinesis. They use this to explain how mitosis helps a child grow from a 3 kg newborn to a 50 kg teenager.
Knowledge	The learner should be able to: a) explain the significance of mitosis and meiosis; b) illustrate the stages of mitosis in cells. Learners recognise that cell division has two types (mitosis and meiosis), and that mitosis has four stages (prophase, metaphase, anaphase, telophase) followed by cytokinesis, producing two daughter cells with the same chromosomes as the parent cell.
Skills	Developing and using models (drawing and interpreting a mitosis diagram); Analysing and interpreting data (observation table of chromosome movement); Obtaining, evaluating and communicating information (matching observations to scientific terms from a reading); Constructing explanations (linking mitosis to growth).
Attitudes	Core competency, Creativity and Imagination: innovatively representing chromosome movement. Core competency, Digital Literacy: where available, watching a mitosis animation. Value, Unity: collaborating in groups and jigsaw teams. Value, Respect: appreciating others' efforts during the gallery walk. PCI, Safety and Security: observing safety precautions around any microscope or slide use.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lessons 1 and 2 established that growth needs many new cells and that chromosomes carry the instructions. Lesson 3 supplies the mechanism: mitosis copies and shares chromosomes accurately so each daughter cell gets the right instructions. It ends with a new question: if mitosis makes cells similar to the parent cell, how are the cells for sexual reproduction (gametes) made? This leads to meiosis in Lesson 4.
Safety Notes	Diagram-based lesson, so low risk. If prepared onion root tip slides and light microscopes are used for demonstration, carry microscopes with two hands, keep slides and cover slips away from table edges, never use direct sunlight with a mirror, and report broken glass to the teacher immediately. Keep walkways clear during the gallery walk.

B. LESSON OVERVIEW

Anchoring phenomenon: a child grows from a 3 kg newborn to a teenager of 50 kg or more, so the body must produce billions of new cells. Learners predict what happens to chromosomes when one cell becomes two. In mixed-ability groups they study a mitosis diagram and record only what they SEE in an observation table (chromosomes appearing, lining up, separating, two nuclei forming, cell dividing) without using stage names. Only then does a short reading give the names, and learners complete 'We observed _____. This is called _____.' A jigsaw shares expertise for large classes, with each expert group handling one stage. A gallery walk and discussion diamond let learners judge claims, and learners return to the phenomenon to explain how mitosis helps a child grow using the chain: one cell → chromosomes accurately distributed → two daughter cells → more cells → growth and repair. Class-size adaptations: groups of 5-6 and a jigsaw so every learner talks, shared diagrams and readings (one per group), a gallery walk that uses wall and desk space, and exit tickets collected by row leaders.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
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Predict Phase  VIDEO: Mitosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mitosis and Cytokinesis (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the phenomenon: a baby photo versus a teenager now, a 3 kg newborn becoming 50 kg. Individually they draw a quick sketch or write 2-3 sentences predicting: 'A single cell in a child's skin or bone divides into two cells. What happens to the chromosomes (the instructions) inside so both new cells can work properly?' They think-pair-share with a neighbour, then predict whether the two new cells will be the same as, or different from, the original cell. Examples: a growing boy in Kisumu getting taller, or a cut knee from a football match in Nairobi healing.	Display or point to the DQB: 'How does the body produce the enormous number of new cells needed for growth?' Say: 'Think about a baby born at 3 kg who becomes a 50 kg teenager in Kericho. Every new cell needs instructions. What happens to the chromosomes when one cell becomes two?' Give 2 minutes of silent individual writing, then WAIT TIME of 10-15 seconds before any pair talks. Pairs share for 2 minutes. Cold-call 4 pairs: 'What did your partner predict?' Record 3-4 predictions on the board, including misconceptions such as 'the chromosomes are split in half' or 'new cells get new, different instructions'. Say: 'Keep these predictions. We will test them with evidence.' In a large class, ask neighbours in rows to pair up, and cold-call by row.	Think-Pair-Share inside a Predict-Observe-Explain (POE) cycle. Individual prediction commits each learner to an idea, pair talk lets every learner speak even in a large class, and the recorded predictions give the class something to test against evidence in the Observe phase.	Oral questions and observation. Look for: learners who mention chromosomes being copied or shared; learners who think chromosomes are halved or new instructions are made; learners who cannot say anything. Tally the three predictions on the board (same, different, unsure) and use them to focus later teaching. Note 4-5 learners to question again in the Explain phase.
Observe Phase  VIDEO: Phases of meiosis I Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mitosis and Cytokinesis Practice Source: CK-12  Search ARES for similar readings	Part 1 (Investigate, 20 min): In groups of 5-6, learners receive one mitosis diagram (a cell with 4 chromosomes, such as an onion root tip cell, shown from start to finish). They look for and record in an observation table: (1) chromosomes becoming visible; (2) chromosomes lining up; (3) chromosomes separating; (4) two new nuclei forming; (5) the cell dividing. Columns: What we see Where the chromosomes are What changed from the last picture. They do NOT write stage names yet. Where equipment allows, a prepared onion root tip slide is viewed in rotation or a short animation is shown to the whole class. Part 2 (Research, 15 min): After the tables are complete, the teacher hands out one short textbook reading per	Say: 'Today you are scientists. Describe only what you can SEE. Please do not write the stage names yet. I will check your tables before the reading.' Circulate and ask: 'Where are the chromosomes in this picture? How is it different from the one before?' and 'How many chromosomes are at each end of the cell?' Give WAIT TIME of 10-15 seconds after each question. Do not hand out the reading until a group's table has all 5 rows filled, and check each group in turn. Then say: 'Use the reading to find the scientific name for each thing you saw.' Cold-call 5 groups to read out one 'We observed _____. This is called _____.' sentence each, one stage per group. Correct errors by asking: 'What in your table supports that name?' Manage a large class by	Predict-Observe-Explain with Jigsaw. Observing first and naming later lets learners build meaning from evidence, so the terms (prophase, metaphase, anaphase, telophase, cytokinesis) become labels for ideas they already formed instead of facts to memorise. Jigsaw makes each learner responsible for one stage and builds interdependence (Unity).	Check the observation table for these indicators: (a) chromosomes described as becoming visible/condensing; (b) lining up at the middle/equator; (c) separating and moving to opposite poles; (d) two nuclei forming; (e) the cell pinching or a cell plate forming to make two cells. Model check: each group's sentences must match the correct stage to the correct observation. Oral questions: 'Which stage did you see chromosomes lining up in the middle?' Record groups needing support, and look for evidence that both daughter cells receive the same number of chromosomes.

	group. Learners match the terms prophase, metaphase, anaphase, telophase, then cytokinesis, to their observations and complete: 'We observed _____. This is called ____.' for each stage. Jigsaw option for large classes: each group member becomes an expert on one stage (including cytokinesis), meets with experts from other groups to check their sentence and drawing, then returns to teach their home group.	giving the reading to the group's reader only (one copy per group), assigning roles (reader, recorder, drawer, reporter, timekeeper), and having the teacher stand in the middle and rotate through zones. For jigsaw, set a 6-minute timer for expert meetings.		
Explain Phase  VIDEO: Phases of meiosis II Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners return to the phenomenon: 'How does mitosis help a child grow?' First, individually, they write a 3-4 sentence explanation with a labelled sketch of the sequence. They then use a Discussion Diamond: each group receives 9 statement cards (for example, 'Mitosis produces two cells with the same chromosomes as the parent cell', 'A baby grows because its cells get bigger only', 'Mitosis helps heal a wound on the knee', 'Chromosomes are shared equally between daughter cells', 'Eating food makes new cells by itself', 'Cytokinesis divides the cytoplasm') and rank them in a diamond from most important to least important for explaining growth. Groups use the sequence: one cell → chromosomes accurately distributed → two daughter cells → increase in cell number → growth and repair. Link to Kenyan contexts: a child growing from infancy on ugali, sukuma wiki and beans, whose food supplies materials and energy while mitosis supplies new cells; a farmer's wound healing; and root tips of maize and beans growing in a shamba.	Say: 'Go back to our child in the photographs. Use today's evidence to explain: how does mitosis help a child grow?' Give 2 minutes silent writing and WAIT TIME of 10-15 seconds before talk. Pose: 'What exactly do the daughter cells receive, and why does that matter?' Then say: 'Rank the statements in your diamond, and be ready to defend your top and bottom cards using evidence from your table.' Cold-call 6 learners: 'Why did you place this card at the top?' Press for evidence: 'Which stage shows that?' Address the misconception that food alone makes cells: 'Food supplies energy and building materials, but what process makes the new cell?' Link to the predictions on the board and ask: 'Which prediction was supported? Which should we change?' For a large class, give each group one set of cards, have groups pair with a neighbouring group to compare diamonds, and select 2-3 diamonds for whole-class discussion.	Discussion Diamond. Ranking statements forces learners to weigh evidence, justify priorities and resolve disagreements, and it surfaces misconceptions (growth only by cell enlargement, food making cells directly). Constructing a causal chain from the evidence turns observations into an explanation of the phenomenon.	Oral questions and written explanations. Indicators of understanding: the learner states that chromosomes are copied and shared equally; that two daughter cells have the same chromosomes as the parent cell; that the number of cells increases; and that this causes growth and repair. Common errors to note: 'the cell splits in half and chromosomes are halved', 'cells just get bigger', 'mitosis makes different cells'. Use a 3-2-1 rubric: 3 = full chain, 2 = partial chain, 1 = missing the idea of chromosome distribution.

Driving Question Board (DQB) Creation  VIDEO: Mitosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The cell cycle and mitosis Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revisit the DQB and sort their sticky notes: they place answers or partial answers beside earlier questions (for example, 'Where do new cells come from?' and 'How does each new cell get the right instructions?'). They add new questions from today, such as 'Do all cells in the body divide by mitosis?', 'Why are babies' cells dividing faster than an adult's?' and 'If mitosis makes identical cells, how are the cells for reproduction made?' A Gallery Walk follows: groups post their observation tables and diamonds around the room, and learners circulate for 6 minutes with sticky notes, leaving one 'I agree because...' and one 'I wonder...' comment on at least two other groups' work, showing respect for others' efforts.	Say: 'Look at our Driving Question Board. Which questions can we now answer, even partly?' Give WAIT TIME of 10-15 seconds, then cold-call 4 learners to move a question to the 'partly answered' section and justify it. Prompt: 'What new question did today's evidence raise for you?' Guide the class toward the key new question: 'If mitosis produces new body cells similar to the original cell, how are the cells used for sexual reproduction produced?' Write it on the DQB as the question for Lesson 4. For the Gallery Walk, rotate groups every 2 minutes with a signal, and in a large class split the room into two walking streams (left side and right side) so walkways stay clear and safe. Remind learners to give kind, evidence-based feedback.	Gallery Walk with a Driving Question Board. Viewing other groups' work exposes learners to alternative representations and strengthens peer critique, while sorting and adding questions keeps the whole unit visibly coherent and links each lesson to the question that drives the next.	Exit ticket (collect by row leaders): (1) Write the four stages in order, plus cytokinesis. (2) In one sentence, explain how mitosis helps a child grow. (3) One question you still have. Success indicators: correct order (prophase, metaphase, anaphase, telophase, then cytokinesis); mention of two daughter cells with identical chromosomes; and a relevant question (for example, about gametes). Sort tickets into 'secure', 'nearly there' and 'needs reteaching' piles for the start of Lesson 4. Observe the quality of gallery notes as evidence of respectful, evidence-based talk.
Model Building Phase  VIDEO: Mitosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mitosis and Cytokinesis (Flexbook) Source: CK-12  Search ARES for similar readings	Learners revise their cumulative model of 'How the body produces the new cells needed for growth'. In their groups, they add a new section to their model from Lessons 1 and 2: a labelled flow diagram showing one parent cell with chromosomes → chromosomes visible (prophase) → lining up (metaphase) → separating (anaphase) → two nuclei forming (telophase) → cell dividing (cytokinesis) → two identical daughter cells. They link this to the child's growth: more cells → growth of tissues such as bone and muscle → repair of cuts or injuries. Learners then mark what they still cannot explain (for example, how reproductive cells are made) with a question mark and a new question for the next lesson. Creative option: use bottle tops, beads or maize and bean	Say: 'Update your model with today's evidence. Each arrow must show what happens to the chromosomes. What stays the same and what changes?' Give WAIT TIME of 10-15 seconds before groups begin. Circulate with these questions: 'Where do you show the chromosomes being copied and shared?' 'How does your model explain the child growing?' 'What can your model not yet explain?' Do model checks with 5 groups: ask each to explain their model in 30 seconds. Then end by saying: 'Our model works for body cells. But if mitosis makes cells similar to the original, how are the cells for sexual reproduction produced? That is our question for the next lesson.' For a large class, assign the roles of drawer, labeller, checker and reporter and spot-check groups by	Cumulative Model Building with peer model checks. Revising a shared model with seeds, drawings or flow diagrams makes learners' thinking visible, lets them test whether the model explains the phenomenon, and shows its limits, which creates a purposeful need for meiosis.	Model check using the indicators: (a) all five steps shown in order; (b) chromosomes drawn accurately, with the same number in both daughter cells; (c) arrows showing the link to the increase in cell number, growth and repair; (d) a stated gap or new question. Oral questions during the 30-second explanations: 'What would go wrong if the chromosomes were not shared equally?' Record each group as Beginning, Developing or Secure.

	seeds to represent chromosomes and move them through the stages on a desk.	zone rather than visiting every group.		
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D. TEACHER REFLECTION

After the lesson, consider: Did learners record true observations before using the stage names, or did some skip ahead? Which group tables were most accurate, and why? Which misconceptions appeared (halved chromosomes, cells only growing bigger, food directly making cells), and were they resolved in the discussion diamond? Did the jigsaw give every learner in a large class a chance to speak? What do the exit tickets show about who can sequence the stages and who can link mitosis to growth? What adjustments are needed for Lesson 4, for example reviewing the order of stages briefly and introducing meiosis through the question about reproductive cells?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed in the mitosis diagram that chromosomes became visible, lined up in the middle of the cell, separated and moved to opposite ends, two new nuclei formed, and the cell divided into two cells.
What did I learn?	We learned that mitosis has four stages, prophase, metaphase, anaphase and telophase, followed by cytokinesis, and that it produces two daughter cells with the same chromosomes as the parent cell.
How does this explain the phenomenon?	We can explain that a child grows because mitosis copies and shares chromosomes accurately, so one cell becomes two identical cells, which increases cell number for growth and repair. We still ask: if mitosis makes similar body cells, how are the cells for sexual reproduction produced?

LESSON 4 (80 minutes (double lesson)): Meiosis and Its Stages: Why Do Sperm and Egg Cells Have 23 Chromosomes?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use a chromosome-number problem, sequencing of unlabelled pictures, a reading and two videos to discover what meiosis is, describe its stages, and explain why gametes have 23 chromosomes. This adds a second kind of cell division to the evidence from mitosis and moves us toward answering how the body produces new cells for growth.
Knowledge	By the end of the lesson the learner should be able to: (a) explain the significance of meiosis in organisms; (b) illustrate the stages of meiosis I (Prophase I, Metaphase I, Anaphase I, Telophase I) and meiosis II (Prophase II, Metaphase II, Anaphase II, Telophase II); (c) explain why sperm and egg cells have 23 chromosomes and how fertilisation restores 46.
Skills	Developing and using models (drawing and sequencing chromosome behaviour); analysing and interpreting data (chromosome-number calculations); constructing explanations from evidence; obtaining, evaluating and communicating information (reading and videos); collaboration in groups. Core competency: Digital Literacy while watching animations of the stages of meiosis.
Attitudes	Unity: collaborating with group members to sequence and label the stages. Respect: appreciating others' efforts and ideas during the gallery walk and discussion diamond. PCI Safety and Security: handling shared materials and devices safely and sharing them fairly.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lesson 3 gave evidence on mitosis as the way body cells are copied for growth and repair. Lesson 4 shows that the body also needs a different division, meiosis, to make gametes with half the chromosomes so that a child inherits 46 chromosomes (23 from each parent). It raises the question for Lesson 5: What would happen to an organism if mitosis and meiosis did not occur?
Safety Notes	No hazardous materials. Keep bags and walkways clear during the gallery walk. Share devices or the projector for videos at a safe volume and screen distance. Handle scissors (if cutting picture cards) with care, or use pre-cut cards. In large classes, assign group roles to keep movement orderly.

B. LESSON OVERVIEW

Anchoring phenomenon: a 3 kg newborn grows into a 50 kg teenager, and each baby began as one fertilised cell (zygote) formed when a sperm met an egg. Learners first face a chromosome-number problem ($46 + 46 = 92?$) and predict how nature avoids doubling every generation. Working in groups (6 to 8 learners per group for large classes, with jigsaw roles), they sequence unlabelled pictures of meiosis, compare them with a reading and two videos (meiosis I and meiosis II), and label the stages. A gallery walk lets groups check each other's sequences. Learners then return to the problem and explain with evidence why gametes have 23 chromosomes. They add to the Driving Question Board (DQB) and revise their class model. Resources: reading [insert reading here]; video of meiosis I [insert link]; video of meiosis II [insert link]; sets of unlabelled meiosis picture cards (one per group); manila paper and markers; the DQB; the class model chart.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners recall the baby photographs and the growth	Display the problem on the board and say: 'Every one of you began	Predict-Observe-Explain (Predict) with Think-Pair-Share: Learners	Oral questions and prediction check. Look for: (1) correct

 VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	phenomenon: every one of us began as a single cell from a sperm and an egg. In groups they receive the problem: 'A human body cell has 46 chromosomes. A sperm cell and an egg cell must combine during fertilisation. What would happen if each reproductive cell also had 46 chromosomes?' Using think-pair-share, each learner thinks alone, then pairs, then the group calculates the chromosome number after fertilisation ($46 + 46 = 92$), notices what happens if this continues over generations (92, then 184, then 368...), suggests how the problem could be solved, and records a prediction on paper: 'We predict that...' with a sketch of how the sperm and egg might differ from a body cell. Kenyan link: groups consider a Kenyan family where a child resembles both mother and father but still has the same number of chromosomes as everyone else in the village.	as one cell. Where did its 46 chromosomes come from?' Give 1 minute of silent think time (WAIT TIME 10-15 seconds after the question before taking any answer). Say: 'Turn to your partner and share your idea in 2 minutes.' Then circulate and cold-call 4 pairs: 'What number did you get after fertilisation, and what does it tell you?' Ask: 'If this happened in every generation, what would happen to the chromosome number in your grandchildren?' Do NOT confirm or correct the predictions. Say: 'Hold on to your prediction; we will test it with evidence today.' For a large class, assign roles (recorder, reporter, materials manager, timekeeper) and have groups of 6 to 8 split into pairs for thinking.	commit to a prediction before seeing the stages of meiosis, which creates a real need to know. Think-Pair-Share gives every learner, including quiet ones in a large class, a chance to voice an idea before group agreement. The calculation makes the 'doubling problem' visible and sets up the idea that gametes must have half the chromosomes.	calculation of 92 chromosomes in at least 80% of groups; (2) groups that mention that the number would keep doubling; (3) at least some groups suggesting that the sperm and egg must carry half (23) each. Note on a checklist the groups that suggest 'halving' and those with misconceptions such as 'the extra chromosomes get thrown away after fertilisation'. Use these to target questions in the Observe Phase.
Observe Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Jigsaw activity. In each home group of 6 to 8, learners take expert roles: Experts A (meiosis I reading and Video 1), Experts B (meiosis II reading and Video 2). Experts meet in expert groups to study their part (reading [insert reading here]; video of meiosis I [insert link]; video of meiosis II [insert link]) and note what happens to the chromosomes. Back in home groups they receive an envelope of unlabelled pictures of the stages of meiosis and: (1) arrange the pictures in the order they think they occur; (2) describe what happens to the chromosomes in each picture (pairing of homologous chromosomes, crossing over,	Say: 'You are now scientists collecting evidence. First, sequence the pictures using only what you can reason from the chromosomes; do not look at the reading yet.' Allow 8 minutes. Circulate and ask without giving answers: 'Where are the homologous pairs in this picture?' and 'How many chromosomes would you count here?' Then say: 'Now check your sequence using the reading and videos.' Play Video 1 (meiosis I) and pause at each stage: 'What are the chromosomes doing now?' (WAIT TIME 10-15 seconds, then cold-call 3 learners per stage). Repeat with Video 2. Say: 'Which picture shows the first division and which	Jigsaw with Predict-Observe-Explain (Observe) and Gallery Walk: Jigsaw makes each learner responsible for part of the evidence, so all must contribute, which suits a large class. Sequencing unlabelled pictures before the reading lets learners test their own reasoning against the scientific reference, which builds understanding of chromosome behaviour rather than memorisation of names. The gallery walk exposes learners to other groups' models, so they can compare sequences and correct errors with peer evidence.	Model checks and oral questions. Observable indicators: (1) the sequence of pictures matches the correct order of the stages after comparison with the reference; (2) labels are correctly placed for Prophase I to Telophase II; (3) the description for Anaphase I says that homologous chromosomes separate, and the description for Anaphase II says that sister chromatids separate; (4) the group identifies two divisions and four cells at the end. Use a checklist during the gallery walk and note common errors such as confusing meiosis I and II, or mixing up homologous chromosomes and sister chromatids. Re-teach in the Explain Phase using the errors

	separation of homologous pairs, separation of sister chromatids); (3) identify where the first and second divisions occur; (4) compare their sequence with the scientific reference from the reading and videos; (5) label the stages: Meiosis I: Prophase I, Metaphase I, Anaphase I, Telophase I; Meiosis II: Prophase II, Metaphase II, Anaphase II, Telophase II. They record the number of chromosomes in the cell at the start, after meiosis I, and after meiosis II (46 to 23 to 23, with each chromosome having two chromatids at the end of meiosis I and one chromatid after meiosis II). They stick the labelled sequence on manila paper for the gallery walk. The Digital Literacy competency is developed as they pause, replay and discuss the animations.	shows the second? Show me with a finger on your paper.' For a large class, use one shared screen, have the materials manager collect and return cards, and have expert groups seated near the screen. Support learners who struggle with reading by pairing them with a strong reader and using the diagrams. Gallery walk instruction: 'Rotate clockwise every 3 minutes. Leave one sticky note with a star (what is correct) and one with a question (what you wonder).' Remind learners of the value of Respect: 'Give feedback kindly and with evidence.'		found.
Explain Phase  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners return to the original problem: 'A human body cell has 46 chromosomes. Why must sperm and egg cells have 23 chromosomes?' Using a discussion diamond, each group ranks four statements on a diamond (most to least convincing, with evidence): (a) Meiosis halves the chromosome number, so gametes have 23; (b) Gametes have 46 chromosomes and the extra are removed after fertilisation; (c) Fertilisation restores 46 chromosomes in the zygote; (d) Meiosis is just the same as mitosis. They justify the rankings with evidence from their sequences, videos and reading. Then each learner writes a short explanation with a flow diagram: Meiosis, then chromosome number is reduced by half, then	Say: 'Return to your first prediction. What do you want to keep, and what do you want to change, based on evidence?' Introduce the diamond: 'Rank these four statements. You must point to evidence from your pictures, the reading or the videos.' Allow 6 minutes of group talk. Then cold-call 5 groups: 'Which statement did you place at the top, and what evidence supports it?' (WAIT TIME 10-15 seconds before each response). Probe: 'How do we know the number is halved at meiosis I and not meiosis II?' and 'Why must the second division follow without copying the DNA again?' Challenge misconceptions: 'If the gametes had 46, what would the zygote have?' Say: 'Write your explanation in your own words with the chain of reasoning.' Circulate	Discussion Diamond with Predict-Observe-Explain (Explain) and Think-Pair-Share: Ranking statements forces learners to weigh claims against evidence and argue why some explanations are stronger, which deepens conceptual understanding. Returning to the original prediction completes the POE cycle and makes learning visible. Individual writing after discussion ensures every learner builds their own explanation of why gametes have 23 chromosomes.	Oral questions and exit ticket. Exit ticket (3 minutes, individual): (1) 'Explain in two sentences why sperm and egg cells have 23 chromosomes.' (2) 'Name the stage in which homologous chromosomes separate.' (3) 'State one difference between mitosis and meiosis.' Indicators of success: the explanation includes the full chain (meiosis, half the chromosomes, 23 in gametes, fertilisation restores 46); the answer to (2) is Anaphase I; the difference is correct (e.g., meiosis gives four genetically different cells with 23 chromosomes, mitosis gives two identical cells with 46). Sort the tickets into three piles (secure, partly secure, not yet) to plan the start of the next lesson.

	reproductive cells (sperm and egg) have 23 chromosomes, then fertilisation restores 46 chromosomes in the zygote, which divides by mitosis to form a growing baby. Learners also compare meiosis with mitosis (number of divisions, number and type of daughter cells, chromosome number) and explain how crossing over helps explain why children in a Kenyan family, for example siblings in Kisumu or Kericho, look similar but not identical. They revisit their first predictions and revise them, stating 'We changed our idea because...'.	and push for the complete chain: Meiosis, then half the chromosomes, then 23 in each gamete, then fertilisation restores 46. Ask learners to compare meiosis and mitosis in a table, with a Kenyan example: 'A runner like Faith Kipyegon grows and repairs muscle using mitosis, but the cells that will make her future children come from meiosis.' For a large class, ask groups to write on manila paper and have reporters give 1-minute summaries.		
Driving Question Board (DQB) Creation  VIDEO: Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board with the main question: 'How does the body produce the enormous number of new cells needed for growth?' In groups they write on sticky notes: (1) one thing they now know about how cells are made (for example, 'Meiosis makes gametes with 23 chromosomes'); (2) one new question that has come up. They place the notes on the DQB under categories such as 'Mitosis', 'Meiosis', 'Chromosomes' and 'Fertilisation', and may link their questions to earlier ones. They read the teacher's highlighted question: 'What would happen to an organism if mitosis and meiosis did not occur?' and discuss in pairs what they think could happen to a growing child, a farmer's maize crop in the Rift Valley, or a fish in Lake Victoria.	Say: 'Take 2 minutes to write one thing we have figured out and one question we still have.' Move around and prompt: 'What puzzles you about how a single cell becomes a whole person?' Invite 4 learners to post and read their notes (WAIT TIME 10-15 seconds before choosing). Group similar questions and say: 'Which of these can we answer with the evidence we have now, and which need new investigations?' Add the new question to the board: 'What would happen to an organism if mitosis and meiosis did not occur?' Ask: 'What might happen to a child's growth, or to reproduction, if neither division happened?' Thank learners for respectful questioning. For a large class, use one board section per group and have a rotating group representative place the notes.	Think-Pair-Share and Gallery Walk of the DQB: Learners connect lesson evidence to the unit's driving question, and see how their questions are valued and drive the next investigations. Categorising notes helps them notice patterns and gaps in their understanding, while the new question creates a purpose for Lesson 5.	Oral questions and observation of DQB contributions. Indicators: at least 80% of groups post a statement that correctly uses evidence from this lesson (for example, gametes have half the chromosomes); questions are linked to growth or cell division, not only recall; learners can name which questions meiosis and mitosis can already answer. Note groups with only factual questions and prompt them to ask 'why' and 'what if' questions.
Model Building Phase  VIDEO:	Groups revise the class model of how the body produces new cells for growth. On a shared chart showing the life cycle (adult body	Say: 'Look at our class model from Lesson 3. What part of it explains the 3 kg baby, and what part is missing?' Give 3 minutes of silent	Model Building with Think-Pair-Share: Building and revising a shared model helps learners combine evidence from different	Model checks. Observable indicators: the model shows meiosis producing gametes with 23 chromosomes; fertilisation is

Fertilization terminology: gametes, zygotes, haploid, diploid Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12 Search ARES for similar readings	cells, then meiosis in reproductive organs, then sperm (23) and egg (23), then fertilisation, then zygote (46), then repeated mitosis, then growth to a teenager), they add new elements from this lesson: meiosis I and II with the stages, the halving of chromosomes, and how gametes join to restore 46. They add labels, arrows and a short key, and write a caption explaining how the model explains the baby-to-teenager phenomenon. Then they do a model check: each group's reporter says what they added, what remains unexplained (for example, how cells know when to stop dividing, or what happens when division goes wrong), and what evidence supports each part. Learners copy the revised model into their notebooks.	time to think (WAIT TIME 10-15 seconds before taking ideas). Then say: 'Add what you learned about meiosis and show where the chromosome number changes.' Circulate and ask: 'Where on your model is the number 23, and where is 46? Show me with your finger.' and 'How does your model explain why every cell in your body has 46 chromosomes?' Cold-call 3 groups for a model check: 'What did you add, and what evidence supports it?' Highlight the unanswered parts and connect them to the new question: 'What would happen to an organism if mitosis and meiosis did not occur?' Say: 'Think about this before the next lesson and write your guess in your notebook.' Remind learners to appreciate each group's effort (Respect). For a large class, assign each group one section of the model to update, then merge into one class model.	lessons into one explanation and see how new knowledge changes it. Identifying gaps shows that scientific explanations grow over time and motivates the next investigation.	shown restoring 46; mitosis is shown as the process by which the zygote grows into a child; the group can state the evidence for each part; and the group identifies at least one unanswered question. Learners who cannot place 23 and 46 correctly should be given a short oral check at the start of Lesson 5.
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D. TEACHER REFLECTION

After the lesson ask yourself: (1) Did every learner, including those in the back rows of a large class, contribute evidence in their group through their jigsaw role? (2) Which stage of meiosis caused most confusion, and did the confusion come from homologous chromosomes and sister chromatids being mixed up? (3) What did the exit tickets show about learners' ability to explain why gametes have 23 chromosomes? (4) Did the discussion diamond expose misconceptions such as 'meiosis is the same as mitosis' or 'extra chromosomes are discarded'? (5) How well did the Kenyan examples (family resemblance, athletes, crops) help learners connect the idea to the growth phenomenon? (6) What will you change next time about grouping, timing or resources, for example more picture sets or a printed copy of the reading for groups without a screen view?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed unlabelled pictures of meiosis and sequenced them, watched videos of meiosis I and II, read about the stages, and calculated that two cells with 46 chromosomes each would give a zygote with 92 chromosomes.
What did I learn?	We learned that meiosis is a division in reproductive organs that has two divisions (meiosis I and meiosis II) and produces four cells, each with half the chromosome number (23). In meiosis I homologous chromosomes separate, and in meiosis II sister chromatids separate.

How does this explain the phenomenon?

We can explain that sperm and egg cells must have 23 chromosomes because meiosis halves the chromosome number, so that when they fuse at fertilisation the zygote has 46 chromosomes again and can then divide by mitosis to grow into a child. Without meiosis the number would double in each generation. We still wonder: What would happen to an organism if mitosis and meiosis did not occur?

LESSON 5 (80 minutes (double lesson: 2 x 40 minutes)): Significance of Mitosis and Meiosis: Case Studies from Growth, Healing, Roots and Offspring

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use real-life case studies to work out why organisms need mitosis and meiosis, and link each case to the correct type of cell division using evidence. This explains where the billions of new cells in a growing child come from and how each new cell gets the right instructions.
Knowledge	a) Explain the significance of mitosis and meiosis. d) Recognise the importance of cell division in reproduction, growth, and development. Mitosis produces two genetically identical diploid daughter cells for growth, repair and asexual reproduction. Meiosis produces four genetically different haploid cells (gametes) for sexual reproduction, which keeps the chromosome number constant and creates variation.
Skills	Analysing case studies; identifying evidence for claims; constructing explanations (NGSS SEP: Constructing Explanations); engaging in argument from evidence; using a table to organise data; communicating ideas to peers.
Attitudes	Unity: collaborates with group members in jigsaw and discussion diamond tasks. Respect: appreciates others' efforts and ideas when presenting and during the gallery walk. Curiosity about how living things such as a Kenyan child, a maize plant or a kienyeji chicken produce new cells and offspring.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lessons 1-4 gave evidence on chromosomes and the stages of mitosis and meiosis. Lesson 5 shifts from how cells divide to why. Learners connect the two processes to growth, healing and reproduction, and so answer the driving question: the body makes new cells by mitosis, each with a full set of identical instructions. The lesson also raises a new puzzle for Lesson 6: if reproduction needs sex cells, how can a single organism produce a new organism without them?
Safety Notes	No hazardous materials. Seat learners so that movement during the gallery walk is orderly in a crowded classroom. Keep bags and stools out of walkways. If the class is large (60+), use groups of 6 and stagger gallery walk rotations to avoid crowding. Use only provided printed case cards. Remind learners to treat photos of classmates and family respectfully (Respect value). No wound-related practicals; the skin healing case is discussed only.

B. LESSON OVERVIEW

Learners return to the anchoring phenomenon: a 3 kg newborn becomes a 50 kg teenager, so the body must make billions of new cells. They first predict which type of cell division explains five scenarios: a growing child, a healing cut, a maize or sukuma wiki plant growing new roots and shoots, a chicken producing chicks, and parents producing children who differ from each other. In home groups of 5-6, learners then use a jigsaw to become experts on one case from a printed reading (the 'reading here' resource, with Kenyan case cards). They return to teach their group, complete a case-study table linking each scenario to mitosis or meiosis with evidence, and check it through a discussion diamond and gallery walk. In the Explain phase learners use think-pair-share to answer the driving question. They add new questions to the Driving Question Board, including how a single organism can produce a new organism without sex cells. They then revise their cumulative class model of cell division with the significance of mitosis (growth, repair, asexual reproduction) and meiosis (gametes, variation, constant chromosome number). Management for large classes: jigsaw groups, shared case cards, one reporter per group, gallery walk in two halves.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners look at the baby photo they brought (or a projected or drawn image of a Kenyan baby and teenager) and recall the 3 kg to 50 kg puzzle. Individually, they write on a scrap of paper a prediction for each of the 5 scenarios: child growing, cut skin healing, plant growing new roots and shoots, chicken producing offspring, parents producing children with different characteristics. They write 'mitosis', 'meiosis' or 'not sure' with a one-sentence reason. They then pair up and compare predictions, noting where they disagree.	Display the five scenarios on the board (or read them aloud for a large class). Say: 'Remember our puzzle: a baby of 3 kg becomes a teenager of 50 kg. Today we find out why cells divide in different ways. Before I give you any evidence, write your best prediction for each scenario. There are no wrong predictions yet.' Give WAIT TIME of 10-15 seconds after each scenario is read before learners write. Say: 'Turn to your partner. Where do you disagree? Why?' Cold-call 4 pairs (from different parts of the room) to share one disagreement. Record the count of learners predicting mitosis vs meiosis for the child growing and children differing from parents using a show of hands on the board. Do not correct any answers. Say: 'Keep your prediction paper. We will return to it at the end.'	Predict-Observe-Explain (Predict step) combined with Think-Pair-Share. Writing predictions first surfaces prior ideas such as 'we grow because of the food we eat' and creates a reason to look at evidence. Pair discussion lets learners hear alternatives, and the disagreement creates a need for evidence in the Observe phase.	Oral questions and observation. Indicators: (1) at least 80% of learners have written a prediction for all 5 scenarios; (2) learners give a reason, not just the word mitosis or meiosis; (3) teacher notes misconceptions heard, e.g. 'the child grows because cells just get bigger' or 'meiosis makes skin cells'; (4) tally of class predictions recorded for later comparison.
Observe Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mitosis vs. Meiosis (Flexbook) Source: CK-12  Search ARES for similar readings	Learners work in jigsaw groups. Home groups of 5-6 are each given five case cards (from the reading). Each member becomes an expert on one case: (A) a child growing from infancy to adolescence, e.g. a baby in Kisumu growing to a teenager in Form 3; (B) a cut on the skin healing, e.g. a boy cut by a panga while weeding shamba; (C) a plant producing new roots and shoots, e.g. a sweet potato vine or a sukuma wiki cutting; (D) a chicken producing offspring, e.g. a kienyeji hen and cock producing chicks; (E) parents producing children with different characteristics, e.g.	Organise the room before starting: groups of 5-6 with 8-10 groups for a class of 50-60, and with several shared case cards (or a board-written copy) where there are not enough printed copies. Say: 'Your job is to be a detective. Each case is a clue to the puzzle of how a baby becomes a teenager. You must give evidence, not just an answer.' Allocate case letters and give 10 minutes for expert groups. Circulate and ask each expert group: 'Which cells are dividing here? Do the new cells need to be identical to the old ones? How do you know?' Give WAIT TIME of 10-15 seconds before prompting.	Jigsaw. Each learner becomes responsible for one piece of the evidence and must teach it, so every learner takes part, which suits a large class. The shared table pushes the group to compare all five cases side by side, so learners can see patterns: mitosis in the cases of growth, repair and asexual reproduction in plants; meiosis in reproduction involving gametes and variation. This links to the earlier lessons on chromosomes and stages of division.	Observation checklist and oral questions. Indicators: (1) expert groups name the correct division for their case; (2) evidence statements are specific (e.g. 'daughter cells identical, same 46 chromosomes' rather than 'because the body needs cells'); (3) all learners in a home group have a complete 5-row table; (4) learners can say why Case E needs meiosis (gametes with half the chromosomes, variation); (5) teacher records groups that mix up the two divisions for follow-up.

	siblings in the same family differing in height or skin tone. Expert groups (all the A's, all the B's, etc.) read their case and answer: 'Which type of cell division is involved, and what evidence supports your answer?' They note evidence such as: identical daughter cells, the same chromosome number, the need for new body cells, gametes with half the chromosome number, and variation among offspring. They then return to their home groups and teach their case. The home group completes the case-study table: Case Type of division Evidence Why it matters to the organism. One member (F, in groups of 6) acts as a checker who asks 'What is your evidence?' for each row.	Prompt for Case E: 'If both parents give a cell, how many chromosomes would a child have if each gave a full set? What must happen first?' Give 15 minutes for home group teaching and table completion. Say: 'Each expert has 2 minutes to teach. Everyone must write in the table.' Cold-call 3 groups to read one row of their table, and ask: 'What evidence did your group use?' Use a timer visible to the class. For large classes, appoint a table monitor per group to give out and collect cards.		
Explain Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mitosis vs. Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners use a discussion diamond to examine claims. Each group receives a diamond of four statements ranked by how strongly they are supported by the evidence, with a middle slot for the most important. Statements: (1) 'Mitosis produces the new body cells that let a child grow.' (2) 'Mitosis replaces damaged skin cells when a wound heals.' (3) 'Meiosis makes gametes so offspring have the correct chromosome number and are different from their parents.' (4) 'Mitosis allows plants to grow new roots and shoots and some plants to reproduce asexually.' Groups rank them and justify the ranking in their own words. Learners then do a think-pair-share on: 'Using your table, explain how a 3 kg baby becomes a 50 kg teenager, and how each new cell gets the right instructions.' They write a 3-4	Say: 'Now you will rank four claims by how strongly the evidence from your cases supports them. You must be able to defend your ranking with evidence from your table.' Give 8 minutes for the diamond. Circulate and ask: 'Which of your cases supports this claim? Could another group rank it differently? Why?' For think-pair-share say: 'Think silently for one minute. Then tell your partner. Then we will share.' Give WAIT TIME of 10-15 seconds before calling on anyone. Cold-call 5 learners, at least 2 who did not speak in the Observe phase, and ask a follow-up of 'What is your evidence for that?' Build on responses, e.g.: 'So a skin cell divides by mitosis and makes two identical cells, each with a full set of chromosomes. That is how each new cell gets the right instructions.' Address a common	Discussion diamond plus Think-Pair-Share, finishing the Predict-Observe-Explain cycle. Ranking forces learners to weigh evidence and justify their reasoning. Think-pair-share gives each learner a chance to build an individual explanation before talking, so quieter learners in a big class participate. Revisiting the predictions helps learners see how their thinking has changed with evidence.	Oral questions and exit ticket. Indicators: (1) learners use evidence from the table to justify the diamond ranking; (2) explanations state that mitosis produces genetically identical daughter cells with the full set of chromosomes; (3) learners distinguish the purpose of mitosis (growth, repair) from that of meiosis (gametes, variation); (4) learners correct the misconception that growth is only cells getting bigger. Exit ticket (written on a half-page): 'Name one situation where mitosis is important and one where meiosis is important. Give one piece of evidence for each.' Collect and sort into 'got it', 'nearly', and 'needs help' piles.

	sentence explanation using the words mitosis, identical, chromosomes and instructions. Finally, they compare their completed table with their Predict-phase paper and note one prediction they changed, completing the Predict-Observe-Explain cycle.	misconception: 'Do we grow only because cells get bigger? What does the evidence from the case of the growing child tell us?' Press learners to connect: 'Where did the extra 47 kg come from? Food provides raw materials, but what makes the new cells?' Clarify that mitosis uses materials from food to build new cells. Conclude with the key ideas: mitosis for growth, repair and asexual reproduction; meiosis for gametes, constant chromosome number and variation.		
Driving Question Board (DQB) Creation  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Kno 100095 Lime Production Traditional Techniques In Sri Lanka Case Study Source: practical_action  Search ARES for similar readings	Learners revisit the Driving Question Board with the driving question: 'How does the body produce the enormous number of new cells needed for growth?' Each group writes on sticky notes or paper strips (1) one thing they now know and (2) one new question. Examples: 'Mitosis makes identical cells for growth and repair' and 'How can a plant grow from a cutting without seeds or flowers?' The class sorts the notes into 'answered', 'partly answered', and 'new'. Learners then walk the board in a gallery walk and add a star to the question they most want investigated next.	Say: 'Look at our driving question. What can we now answer with evidence? What new questions has today raised?' Give 2 minutes for group writing and WAIT TIME of 10-15 seconds before asking for ideas. Cold-call 4 groups to read one note aloud and place it on the DQB. Move the answered questions to the 'answered' column with a cited piece of evidence. Plant the lesson question if learners have not asked it: 'A single sweet potato vine cutting can grow into a whole plant. A bacterium splits into two. No sex cells are involved. How can a single organism produce a new organism without producing sex cells?' Write it on the board as the new question and record it on the DQB as the bridge to Lesson 6. For large classes, one representative per group places the notes while others watch, and the gallery walk is done in two halves so the board is not crowded.	Gallery walk around the DQB. Learners view and respond to the whole class's thinking, which shows how the evidence is connecting to the driving question and points to what the class still needs to find out. Starring questions gives learners a voice in what comes next and builds ownership of the inquiry.	Oral questions and observation. Indicators: (1) every group contributes at least one 'now I know' note and one new question; (2) 'now I know' notes cite evidence from the case-study table; (3) at least one question points to asexual reproduction or budding and binary fission; (4) learners place notes in appropriate categories and justify the choice.
Model Building Phase	Learners revise the class model of cell division using the earlier model of chromosomes and the stages of	Say: 'Open your model from the last lesson. What can we add now that explains how a baby becomes	Cumulative model building with peer model checks. Learners combine evidence from all five	Model checks and exit tickets. Indicators: (1) model labels mitosis for growth, repair and shoots/roots,

 VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  PDF: Kno 100089 Lime Production Traditional Batch Techniques In Patarra Costa Rica Case Study Source: practical_action  Search ARES for similar readings	mitosis and meiosis. Each group adds to a model poster (drawn on manila paper or a double page in their books) the following: a growing child showing one cell dividing by mitosis into two identical cells; a healing wound with mitosis arrows; a plant shoot with mitosis; and a parent producing gametes by meiosis, with a fertilised egg restoring the full chromosome number (e.g. 46 in humans: 23 + 23). They label each arrow with 'mitosis' or 'meiosis' and add a short note on why it matters. Groups then swap with a neighbouring group for a model check, using a checklist of: correct division labelled, chromosome numbers shown, purpose stated, link to the growth phenomenon. Learners add a question mark on the model where they still cannot explain how a single organism reproduces without gametes.	a teenager?' Show the checklist on the board and say: 'Check the other group's model against these four points. Give one star (something done well) and one wish (something to improve).' Give 8 minutes to build and 4 minutes to check, with WAIT TIME of 10-15 seconds when asking groups to justify a label. Circulate and ask: 'Why is this arrow mitosis? How many chromosomes does each daughter cell have? What happens to the chromosome number in meiosis?' Cold-call 3 groups to explain one part of their model in 30 seconds. Use the gaps to prepare Lesson 6. Close with an exit ticket: 'In 2 sentences, explain how the body makes billions of new cells for growth and why meiosis is also needed.' Collect for review. Say: 'Next lesson we investigate how a single organism, like a bacterium or yeast, can produce a new organism without sex cells.'	cases into one visual model, which helps them connect the processes and the purposes of division. Peer checks against clear criteria help learners spot gaps and use scientific vocabulary. Marking the unresolved question on the model carries the inquiry into Lesson 6.	and meiosis for gamete formation; (2) model shows chromosome numbers (e.g. 46 to 46 in mitosis; 46 to 23 in meiosis) with fertilisation restoring 46; (3) learners state purpose (growth, repair, variation) on the model; (4) peer feedback uses the checklist; (5) exit ticket accurately states that mitosis produces identical cells for growth and that meiosis gives gametes and variation. Teacher sorts exit tickets into three piles to plan Lesson 6 starter.
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D. TEACHER REFLECTION

Which groups could name the division but not give evidence, and what prompts helped them? Did learners still say that a child grows only because cells get bigger or because of food alone? Did the jigsaw give every learner in the large class a chance to speak, or did some rely on group leaders? Which case (child, skin, plant, chicken, siblings) caused the most confusion, particularly the Case E link to variation? How well did learners connect chromosome numbers (46 to 46 in mitosis, 46 to 23 in meiosis) to purpose? Were the Kenyan contexts (panga cut, sweet potato vine, kienyeji chickens) meaningful to learners? Which exit tickets show learners are ready for Lesson 6 on budding and binary fission, and who needs extra support?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Case studies of a growing child, a healing skin cut, a plant producing new roots and shoots, a chicken producing offspring and parents producing children with different characteristics; for each we looked at evidence such as identical or different cells, chromosome numbers and the purpose of the new cells.
What did I learn?	Mitosis produces genetically identical diploid cells for growth, repair and asexual reproduction in plants, which is how a child makes billions of new cells with the right instructions. Meiosis produces haploid gametes with different combinations of genes, so

	fertilisation restores the chromosome number and offspring vary from their parents.
How does this explain the phenomenon?	A 3 kg baby becomes a 50 kg teenager because body cells divide repeatedly by mitosis, using materials from food, with each new cell receiving an identical full set of chromosomes. Meiosis is used only to make gametes for sexual reproduction. New question: How can a single organism produce a new organism without producing sex cells?

LESSON 6 (80 minutes (double lesson)): Budding and Binary Fission: Making New Cells and Organisms from One Parent

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners compare budding in yeast and binary fission in bacteria, model both processes, present the main differences, and conclude that both are asexual reproduction from a single parent without fusion of sex cells. This advances the driving question by showing one way cell division produces new cells and organisms with the same instructions.
Knowledge	c) Distinguish between budding and binary fission in relation to cell division. d) Recognise the importance of cell division in reproduction, growth, and development. Learners state that budding and binary fission are asexual reproduction in which a new organism arises from a single parent without fusion of sex cells.
Skills	NGSS SEP: Developing and using models (build and revise physical models of budding and binary fission); Analyzing and interpreting data (compare observations in a table); Constructing explanations (write evidence-based claims); Engaging in argument from evidence (defend differences in the gallery walk). Core competency - Creativity and Imagination: the learner innovatively models budding in yeast and binary fission in bacteria.
Attitudes	Unity: the learner collaborates with others while modelling and in jigsaw groups. Respect: the learner appreciates others' efforts while presenting and in the gallery walk. PCI - Communicable Diseases: the learner recognizes the importance of healthy practices (handwashing, boiling drinking water, food hygiene) for preventing communicable diseases while modelling bacterial fission.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	In Lessons 1-5 learners gathered evidence on chromosomes, the need for new cells, and mitosis. Lesson 6 shows that single-celled organisms use cell division to reproduce, so one parent can make a new organism. It links to the growth phenomenon: just as one yeast cell becomes a colony, a single fertilised cell becomes a 50 kg teenager through repeated division. It raises the next question: how do the mechanisms and types of binary fission and budding differ in detail (DifferenceBetweenBinaryFissionandBudding_MechanismTypesComparison)?
Safety Notes	Do NOT open or taste any yeast or bacteria cultures. Use only sealed dried baker's yeast, in a sealed or covered container, and model bacteria with plasticine, paper or drawings, not real cultures. Wash hands with soap and water before and after the lesson. Wipe benches with disinfectant. Keep the sugar-yeast mixture away from the face. Large class: use groups of 6 and keep walkways clear for the gallery walk.

B. LESSON OVERVIEW

Learners begin by predicting how a single organism can make a new organism without sex cells. Using the phenomenon of a baby growing to a 50 kg teenager, they link this to how cells multiply. In groups of 6 (suited to a large class) they work as a jigsaw: expert pairs study yeast (as in mandazi dough rising or busaa brewing), bacteria (as in Escherichia coli or Salmonella in contaminated water) or a comparison table. They build models with plasticine, paper and string, complete a comparison table, take part in a discussion diamond to rank differences, and present in a gallery walk. They end by writing an evidence-based explanation that both are asexual reproduction from a single parent without fusion of sex cells, add to the Driving Question Board, and revise the class model.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual Reproduction Practice Source: CK-12  Search ARES for similar readings	Learners recall the baby-photo and time-lapse phenomenon of a 3 kg newborn becoming a 50 kg teenager. They look at a pictures of a rising mandazi dough or a bubbling bowl of yeast, and a diagram of bacteria in a contaminated water sample from a Kisumu lakeshore. Individually they write a prediction on a half page: 'How can a single yeast cell or bacterium produce a new organism without sex cells? What do you think happens to its DNA and its cell body?' They sketch a quick prediction drawing, then think-pair-share.	Display the baby photos and say: 'A single yeast cell in this dough becomes millions in two hours. How could one cell do that with no mother and father?' Give silent individual WAIT TIME of 10-15 seconds, then 3 minutes to write and draw. Say: 'Turn to your partner and share your prediction, and listen for one idea that differs from yours.' Cold-call 4 pairs from different parts of the room (front, middle, back, side) and record predictions on the board without judging. Ask: 'What word would you use for making offspring from one parent?' WAIT TIME 10 seconds. Circulate and note common ideas such as 'the cell splits' or 'it needs a partner'.	Think-Pair-Share and Predict-Observe-Explain. Learners commit to a prediction so they can later compare it with evidence. Sharing with a partner brings alternative ideas out in a large class where everyone cannot speak whole-class, and the recorded predictions give a baseline for revision in the Explain phase.	Oral questions and prediction sketches. Look for: learners who mention one parent, splitting or copying DNA (ready to build on); learners who say two parents or sex cells are needed (misconception to address). Tally on a clipboard how many of the 4 cold-called pairs mention splitting, DNA copying or bud-like growth.
Observe Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Understand: sexual and asexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In jigsaw groups of 6, each home group splits into three expert pairs. Pair A (Yeast): observe a picture or teacher-prepared sealed yeast-and-sugar bottle with a balloon, plus a diagram of budding, and model a mother yeast cell with plasticine, making a small bud that grows, receives a copy of the nucleus (a bead), and separates. Pair B (Bacteria): use a card diagram of binary fission and model a rod-shaped bacterium with plasticine and string (the circular chromosome), copying the string, elongating the cell, forming a dividing wall and splitting into two equal cells. Pair C (Comparison): prepare a table with the headings: parent cell, how the new cell forms, size of daughter cells, number of cells formed, organism example, and whether sex cells are involved. After 15 minutes the experts return to their	Distribute pre-prepared materials by group to reduce movement in the big class. Say: 'Wash your hands now. No one opens the yeast bottle. We model bacteria, we do not grow them.' Circulate and ask: 'Where is the copy of the DNA in your model?' and 'Is the bud the same size as the parent?' Use WAIT TIME of 10-15 seconds after each question. Remind groups: 'Bacteria are not always harmful, but some cause cholera and typhoid, so we boil drinking water and wash hands.' Cold-call 3 experts per type (9 total) during the teach-back phase to check accuracy. Give time cues: 'Five minutes left before the experts return.' Prompt the final data question: 'How many bacteria after 2 hours if one divides every 20 minutes? Show the doubling.' Allow 10-15 seconds of WAIT TIME before taking answers.	Jigsaw activity with modelling. Each expert becomes responsible for one piece of evidence, which promotes interdependence and Unity, while building physical models makes the invisible process of cell division concrete. Teaching others forces learners to articulate the mechanism, and the doubling calculation links directly to the exponential increase in cell numbers in a growing child.	Model checks. Observable indicators: the yeast model shows an unequal bud and a nucleus copy moving into it; the bacterial model shows the chromosome copied before the cell splits into two equal parts; the table has at least four correct differences; the doubling sequence 1, 2, 4, 8, 16, 32, 64 is correct. Use a checklist during circulation and mark 'secure', 'developing' or 'not yet' for each group.

	home groups and teach the others, and the group completes the comparison table. Learners then count: if one bacterium divides every 20 minutes, how many after 2 hours? They record 64 cells and relate this to how fast cell numbers rise.			
Explain Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Asexual vs. Sexual Reproduction Practice Source: CK-12  Search ARES for similar readings	Groups take part in a discussion diamond. Each group receives nine statements on cards, such as: 'Budding forms unequal cells', 'Binary fission forms two equal cells', 'Both need only one parent', 'Both involve fusion of sex cells', 'Both copy genetic material before dividing', 'Budding occurs in yeast', 'Binary fission occurs in bacteria', 'Daughter cells are genetically identical to the parent', and 'These processes help fermentation of busaa and rising of dough'. They rank them in a diamond from most important to least important for answering the question 'How can a single organism produce a new organism without producing sex cells?' Groups then link back to the question and each learner writes a short explanation using evidence from the models, the table and the doubling data, arriving at the idea that budding and binary fission are forms of asexual reproduction in which new organisms arise from a single parent without fusion of sex cells. They also compare with their original prediction and correct it.	Say: 'Place the card you believe is the strongest evidence at the top of the diamond. Be ready to justify it with evidence from your model.' Give 8 minutes and circulate. Ask: 'Which card would you remove because it is false?' (the sex cell card) WAIT TIME 10-15 seconds. Cold-call 5 groups to defend their top card and 2 groups to explain the false card. Then give the sentence frame: 'Budding and binary fission are ____ because ____, shown by ____.' Allow 5 minutes of writing. Ask: 'How is this like what happens when a baby grows? How is it different?' WAIT TIME 10-15 seconds, then guide: growth in a child uses mitosis to add cells to one body, while here the new cells become new organisms. Reveal the correct term 'asexual reproduction' only after learners have offered their own words.	Discussion diamond. Ranking forces learners to weigh evidence, justify their decisions and see that different groups prioritise differently, which deepens reasoning. Connecting this to the child-growth phenomenon helps them see cell division as a shared mechanism for both reproduction and growth.	Oral questions and written explanations. Success indicators: explanation states one parent; no fusion of sex cells; offspring are genetically identical; mentions DNA copying before division; names yeast for budding and bacteria for binary fission. Collect 10 written explanations at random from a large class and sort into 'complete', 'partial' and 'misconception' piles to plan the next lesson.
Driving Question Board (DQB) Creation  VIDEO: Asexual and sexual reproduction	Learners hold a gallery walk. Each group posts its model, comparison table and a poster titled 'Budding vs Binary Fission' around the classroom. Groups rotate every 3 minutes, leaving a sticky note with one 'I notice...' and one 'I wonder...' comment, showing	Say: 'Walk with your group, read respectfully, and leave one kind, specific comment. Use the phrase I notice or I wonder.' Time each rotation with a bell, 3 minutes each, with 4-5 stations to manage a large class (two groups per station if needed). Afterwards ask:	Gallery walk. Moving between displays exposes learners to many models and ideas without needing whole-class presentations, supports peer feedback and helps them notice patterns across groups. Posting questions on the DQB keeps the class inquiry	Exit tickets and peer comments. Observable indicators: sticky-note comments refer to specific features (e.g. 'your bud is smaller than the parent'); questions move from recall to mechanism or comparison; at least 3 new questions link to the growth

Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Asexual Reproduction Practice Source: CK-12 Search ARES for similar readings	respect for others' work. Back at their desks, each learner writes one new question on a sticky note, for example: 'Do plants or animals also reproduce by budding?' (Hydra), 'How is the genetic material shared in unequal budding?' or 'How does cell division in a human child differ from fission?' and posts it on the Driving Question Board under the headings 'How cells divide', 'Where cells come from' and 'How instructions are copied'. They add the chart on the DQB: 'Budding and binary fission = asexual reproduction, one parent, no sex cells.'	'What new question has this raised for you?' WAIT TIME 10-15 seconds. Cold-call 4 learners to read their question aloud and place it on the DQB. Sort questions with the class: 'Which questions can we answer from our evidence so far, and which need more investigation?' Highlight the question that leads to the next lesson on detailed mechanisms and types of binary fission and budding (DifferenceBetweenBinaryFissionandBudding_MechanismTypesComparison).	visible and tied to the driving question.	phenomenon. Exit ticket: 'Write one difference between budding and binary fission and one similarity.' Collect at the door and check at least 80 percent correct.
Model Building Phase VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Asexual Reproduction Practice Source: CK-12 Search ARES for similar readings	Learners return to the class model of 'How the body makes enough new cells for growth'. In their notebooks they revise their individual model by adding a new panel titled 'Single-celled organisms: one cell becomes two'. They draw yeast budding and bacterial fission with labels (parent cell, bud, chromosome copy, daughter cells) and write an arrow note showing how the idea links with the growth of a child from 3 kg to 50 kg: 'Dividing cells increase in number, copying instructions each time.' They also note one health link: bacteria multiply quickly in unclean water, so boiling water and handwashing prevent diseases such as cholera and typhoid. Finally they complete the summary table (Observed, Learned, Explained).	Say: 'Open your model from Lesson 5. Add a new panel and show what is the same and what is different when a single cell divides to form a new organism.' Circulate and ask: 'Where in your model do the instructions get copied?' WAIT TIME 10-15 seconds. Cold-call 3 learners to explain one revision and 2 to describe the health link. Close with: 'What is still unclear about how a cell copies its instructions accurately?' Collect 5 notebooks as model checks and note which ideas need to be reinforced. Preview the next lesson.	Cumulative model revision. Adding the new panel lets learners integrate new evidence into their prior understanding of mitosis and growth, make cause-and-effect links, and identify gaps that motivate the next lesson.	Model checks and oral questions. Indicators: model includes a chromosome or DNA copy before division; labels bud and daughter cells correctly; shows one parent only; links to growth of the child; states a health practice. Use a 3-level rubric (complete, partial, missing) and record counts to plan reteaching.

D. TEACHER REFLECTION

Which groups' models showed the DNA being copied before division, and which skipped that step? Did the large class size limit access to materials or participation in

the jigsaw and gallery walk, and how could I rearrange groups or stations? Which learners still believe sex cells are needed for all reproduction? Were the Kenyan contexts (mandazi, busaa, contaminated water) helpful, and did learners link single-celled division to growth of a child? What questions on the DQB should drive the next lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We observed that yeast produces a small bud that grows and separates from the parent, while a bacterium copies its chromosome, elongates and splits into two equal cells. A single bacterium dividing every 20 minutes gives 64 cells after 2 hours.
What did I learn?	We learned that budding gives unequal parent and daughter cells (yeast), binary fission gives two equal cells (bacteria), and both involve copying genetic material before the cell divides. Both are fast ways to increase cell numbers.
How does this explain the phenomenon?	Budding and binary fission are forms of asexual reproduction in which a new organism arises from a single parent without fusion of sex cells, so the offspring are genetically identical to the parent. This shows how cell division produces new cells, as in the growth of a child, and raises the question of how the mechanisms and types compare in more detail.

LESSON 7 (80 minutes (double lesson)): Microscope Practical on Mitosis: Catching Growth in Action in Onion Root Tips

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners prepare and observe onion root tip slides with a light microscope, following safety precautions, to identify stages of mitosis and link what they see to how a child grows from a 3 kg newborn to a teenager of 50 kg or more.
Knowledge	By the end of the lesson, the learner should be able to: (a) explain the significance of mitosis; (b) illustrate the stages of mitosis in cells; (d) recognise the importance of cell division in reproduction, growth and development. Learners know that growing tissues such as root tip meristems contain many cells at different stages of mitosis, and that each stage has a recognisable chromosome pattern.
Skills	Developing and using models (drawing what is seen under the microscope); planning and carrying out investigations (preparing a root tip squash); analysing and interpreting data (counting cells per stage and calculating the proportion in each stage); constructing explanations; engaging in argument from evidence; communicating findings to peers.
Attitudes	Core competencies: Creativity and Imagination (innovatively modelling and representing the stages of cell division); Digital Literacy (using a short animation to check observations). Values: Unity (collaborating in groups while using the light microscope); Respect (appreciating others' efforts and drawings during the gallery walk). Learners show curiosity, patience and care with equipment.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lessons 1 to 6 built the idea that growth needs many new cells and that cells carry instructions in chromosomes. Lesson 7 provides the first direct visual evidence: dividing cells can actually be seen in a growing tissue. Learners see chromosomes behaving in distinct stages, which supports the claim that new cells are made by division of existing cells, not from food directly. This evidence feeds Lesson 8 (meiosis and the contrast with mitosis) and the final model.
Safety Notes	PCI Safety and Security: Carry the microscope with two hands, one under the base and one on the arm. Never use direct sunlight as a light source. Always start focusing on low power and use only the fine focus on high power to avoid breaking slides. Handle slides and cover slips by the edges and dispose of broken glass in the sharps container. The stain (acetic orcein or toluidine blue) stains skin and clothes, so wear gloves, aprons and eye protection. Hydrochloric acid (1M) is an irritant: the teacher dispenses it, and any spill is rinsed with plenty of water. Use a hot water bath, not a naked flame, for warming roots. Wash hands after the practical. Where the school lacks enough microscopes, use prepared slides or photomicrographs so that all learners can take part safely.

B. LESSON OVERVIEW

Learners return to the puzzle of how a newborn of about 3 kg becomes a teenager of 50 kg or more. They predict where in a growing plant or body new cells are made, then prepare and observe onion (kitunguu) root tip squashes in small groups, rotating at the microscope because of the large class size. Using a jigsaw, each group becomes expert in one stage of mitosis and teaches others. Learners tally the cells in each stage, discuss what the numbers suggest about how fast tissues grow, and add their evidence to the Driving Question Board and the class model of growth. Sensemaking strategies: Predict-Observe-Explain, Think-Pair-Share, Jigsaw, Discussion Diamond, Gallery Walk. Formative assessment: oral questions, model checks and exit tickets.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Mitosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners look again at their baby photograph and the time-lapse image of a child growing over 14 years displayed on the board. The teacher shows an onion (kitunguu) with roots growing in a bottle of water, grown for a week, and a root grown in the dark that has lengthened by several centimetres. Learners individually draw a quick sketch in their notebooks predicting: (1) which part of the root they think is making new cells (tip, middle, or the whole root), and (2) what they think the inside of a cell that is making copies will look like under the microscope. They then use Think-Pair-Share to compare predictions with a partner and share with the class.	Hold up the onion with roots and say: 'This root grew 3 cm in five days. Just like you grew from a 3 kg baby. Where do you think the new cells are being made?' Give WAIT TIME of 10-15 seconds in silence before anyone talks. Then say: 'Draw your prediction first, then turn to your partner.' Circulate and listen to the pairs. Cold-call 4 pairs, ensuring at least 2 different predictions are heard, and record them on the board under 'Our Predictions'. Ask: 'What would make you change your prediction?' Remind learners that the safety rules will be reviewed before touching any equipment. With a large class, ask pairs sitting in the same row to share with each other first, so more voices are heard in less time.	Predict-Observe-Explain (Predict) and Think-Pair-Share: Committing to a prediction in writing before the practical creates a cognitive stake, so the later observation either confirms or challenges learners' ideas. Pair talk gives every learner a chance to rehearse ideas, which is vital in a large class where not everyone can speak to the whole group.	Oral questions and observation of sketches. Look for: learners who place new cell production in the root tip or growing region versus those who say the whole root or 'the food becomes cells'. Indicators: at least 2 of every 5 pairs can give a reason for their prediction (e.g. 'the tip is the part that gets longer'). Note misconceptions such as 'cells just get bigger' for later follow-up.
Observe Phase  VIDEO: Phases of meiosis I Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The cell cycle and mitosis Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Safety first: learners review the safety rules and the teacher demonstrates correct microscope handling. In groups of 5 or 6 (each group has one microscope and one prepared root tip kit), learners carry out the following steps: (1) cut about 2 mm from the tip of an onion root grown for a week; (2) place in 1M hydrochloric acid warmed in a water bath for a few minutes to soften the tissue; (3) rinse and place on a slide; (4) add a drop of acetic orcein stain and wait 2 to 3 minutes; (5) cover with a cover slip, cover with filter paper and gently press with a thumb (squash) without sliding the cover slip; (6) view under low power, then high power. Roles are rotated: handler, focuser, recorder, timekeeper, reporter. While one learner is at the microscope, the	Say: 'Hands up if you can name one safety rule before you start.' Cold-call 3 learners to state rules (two hands to carry, low power first, no sunlight, gloves on). Demonstrate the squash technique once at the front, then circulate. Ask each group: 'What do you notice in the cells near the tip compared with the cells further from the tip?' Give WAIT TIME of 10-15 seconds before prompting. Do not tell learners which stage they are looking at; ask 'What are the chromosomes doing? Where are they in the cell?' Visit every group at least once, and ask groups with working slides to help the others (Unity). Use a 'check-in' signal: groups raise a green card when they have a clear field of view so you can verify at least one cell before they continue. For	Predict-Observe-Explain (Observe): Learners generate first-hand evidence that can test their predictions. Rotating roles and micro-tasks (tally, drawing, photomicrograph comparison) ensure that all learners engage with the evidence even when the microscopes are limited by class size. Structured drawing and tallying move learners from 'looking' to 'data'.	Observation checklist while circulating. Indicators: (1) correct handling (two hands, low power first, fine focus only on high power); (2) squash slide shows a single layer of cells; (3) the drawing shows at least 3 distinguishable stages with chromosomes drawn as thread-like or condensed structures; (4) tally chart filled with at least 50 cells; (5) group members take turns and respect each other's contributions. Record names of learners needing support. Use oral questions such as 'Which cell is in metaphase? How do you know?'

	others prepare a second slide, draw observations or study a labelled photomicrograph, so everyone is productively engaged. Learners draw at least 4 different cells showing different appearances of chromosomes and tally 50 to 100 cells by stage (interphase, prophase, metaphase, anaphase, telophase). Where microscopes are short, groups use prepared slides or photomicrograph printouts in the rotation.	groups waiting, give photomicrographs of dividing cells from onion root tips. Announce time: 'You have 5 minutes left to complete your tally.'		
Explain Phase  VIDEO: Phases of meiosis II Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Jigsaw: Home groups number their members 1 to 5. Each number becomes an expert on one stage (1 = interphase, 2 = prophase, 3 = metaphase, 4 = anaphase, 5 = telophase and cytokinesis). Experts from different home groups meet in expert groups, compare their drawings with a labelled diagram or a short animation (if a projector or phone is available), and agree on 3 key features of the stage and one clue to identify it. They then return to their home groups and teach the others, and the home group arranges the stage drawings in the correct order to make a cycle. Next, a Discussion Diamond: groups receive 4 statements and rank them from most to least supported by the evidence: (a) 'The root tip makes new cells by dividing existing cells'; (b) 'Most cells in the tip are dividing at the same moment'; (c) 'Each new cell receives a copy of the chromosomes'; (d) 'A child grows only because the cells get bigger'. Groups justify their ranking using their tally results. Learners then link back to the phenomenon: just as the onion root tip has many	Say: 'You are now experts. Your job is to teach your stage in 2 minutes, not to read your notes.' Set a visible timer. During the expert groups, ask: 'What is the one thing that tells you this is anaphase and not metaphase?' Give WAIT TIME of 10-15 seconds. Cold-call 5 learners from different home groups to explain their stage to the class in one sentence. During the Discussion Diamond, circulate and challenge: 'Which of your observations supports statement (d)? Which does not?' Highlight that if most cells are in interphase, it means mitosis is short compared with the whole cell cycle. Ask: 'If a Kisumu schoolchild grows 6 cm in a year, what would you expect to find in the growth regions of the bones?' Cold-call 3 learners. Use the class tally from each group, written on the board, to show the combined data, and ask: 'What does this tell us about which stage lasts longest?' Correct the misconception that food turns directly into cells: 'Food gives the materials and energy; the cells are made by division of existing cells.'	Jigsaw and Discussion Diamond: Jigsaw makes every learner responsible for teaching a part of the whole process, so understanding is distributed and verbalised, which suits a large class. The Discussion Diamond forces learners to weigh claims against evidence and justify their reasoning, linking the stages observed to the idea that mitosis produces new cells for growth.	Oral questions and observation of Jigsaw teaching. Indicators: learners correctly describe chromosome position in their stage (e.g. 'chromosomes line up at the equator in metaphase'; 'sister chromatids are pulled apart in anaphase'); the diamond ranking places statement (a) and (c) high and (d) low with an evidence-based justification; at least 4 of 5 cold-called learners link the root tip observation to growth in children.

	cells in mitosis, a growing child's tissues (bone ends, skin, blood-forming marrow) make billions of new cells.			
Driving Question Board (DQB) Creation VIDEO: Mitosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12 Search ARES for similar readings	Learners return to the Driving Question Board with the question 'How does the body produce the enormous number of new cells needed for growth?' Each group writes on sticky notes (1) one question that has been answered by today's evidence, and (2) one new question that has arisen (e.g. 'How do the chromosomes get copied before division?', 'Why do some cells divide faster than others?', 'What happens when cells divide without control, like in cancer?', 'Do human cells look the same as onion cells when dividing?'). They place the notes on the board under headings 'Answered', 'Partly answered' and 'New questions'. Learners then do a Gallery Walk of other groups' drawings and tally charts, leaving one comment 'I appreciate...' and one 'I wonder...' on each (Respect).	Say: 'Take 2 minutes to write one question we can now answer and one new question that today's slides made you ask.' Give WAIT TIME of 10-15 seconds after each prompt. Cold-call 4 groups to read one note aloud and ask the class: 'Which question should we investigate next and why?' Cluster questions on the board; point out connections to upcoming lessons (meiosis, uncontrolled division, tissue culture). During the Gallery Walk, set a 6-minute timer, rotate groups clockwise every 2 minutes, and remind learners to give respectful, specific feedback. With a large class, split the room into two galleries so that groups are not crowded around posters. Say: 'Look for one drawing that is different from yours and tell the group why.'	Gallery Walk with the Driving Question Board: Viewing peers' evidence lets learners compare data and drawings, notice variation and strengthen their own ideas. The DQB makes the growth of understanding visible and shows learners that new questions are a product of evidence.	Oral questions and analysis of sticky notes. Indicators: at least half of the groups write a question that moves beyond description, such as asking how chromosomes are copied or how division is controlled; feedback comments are specific and respectful; learners can point to evidence on another group's poster to support or challenge their own findings.
Model Building Phase VIDEO: Mitosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12 Search ARES for similar readings	Learners revise the class and individual models of 'How a child grows from 3 kg to 50 kg'. In their groups, they add today's evidence: a labelled cycle diagram showing a parent cell with chromosomes going through prophase, metaphase, anaphase and telophase to form two identical daughter cells, plus an arrow showing that repeated division (1 to 2 to 4 to 8...) leads to billions of cells. They label where in a child's body they expect rapid mitosis (bone growth regions, skin, bone marrow, gut lining) and add a note on what remains unanswered.	Say: 'Take out your model from Lesson 6. What do you need to add, change or remove after seeing real dividing cells?' Give WAIT TIME of 10-15 seconds. Circulate for a model check: ask each group 'Where does your model show the chromosomes being shared equally?' and 'How does your model explain the billions of cells?' Cold-call 3 groups to explain one revision they made and why. Encourage a calculation: 'If one cell divides every 24 hours, how many cells after 10 days?' (1024). Link to growth: 'Now explain to your partner why a	Model building and revision: Learners integrate new evidence into a causal model (growth results from repeated mitosis of existing cells, which pass identical chromosome copies to each daughter cell), making their reasoning explicit and revisable. Linking back to the phenomenon ensures the practical is not an isolated activity.	Model checks and exit tickets. Indicators: model shows chromosomes duplicated and shared equally between two daughter cells; the stages are correctly ordered; the model explicitly links repeated division to the increase in cell number and body mass; the exit ticket correctly identifies metaphase, explains growth by cell division in one clear sentence and names a valid safety precaution. Learners who cannot do this get a short re-teaching with photomicrographs in the next lesson.

	Each learner then completes an exit ticket individually.	Kenyan child who eats well grows more than one who does not' (materials and energy for division). Distribute the exit ticket: '1. Draw and name the stage in which chromosomes line up at the middle of the cell. 2. In one sentence, explain how what you saw in the onion root tip helps explain how a baby grows into a teenager. 3. Name one safety precaution you followed today and why.' Collect at the door.		
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D. TEACHER REFLECTION

After the lesson, consider: Did all learners get enough time at the microscope given the class size, and which rotation strategies worked best? Which stages were most difficult to identify (often prophase versus early metaphase), and what additional models or photomicrographs could help? Did the tally results show most cells in interphase, and did learners interpret that correctly? Which misconceptions remain (e.g. 'food turns into cells', 'cells only grow bigger')? Did the safety routines hold, and was the stain and acid handled correctly? Which exit ticket responses suggest that learners need more support before moving on to meiosis in Lesson 8?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We prepared onion root tip squashes and, using a light microscope, saw cells at different stages of mitosis (interphase, prophase, metaphase, anaphase, telophase) in the growing tip. Most cells were in interphase, and a smaller number were dividing.
What did I learn?	Mitosis is how existing cells divide to form two identical daughter cells, each with a full set of chromosomes. Growing tissues have many cells in division. Safety precautions are essential when using microscopes, stains and acids.
How does this explain the phenomenon?	A child grows from about 3 kg to 50 kg or more because cells in growing tissues divide by mitosis again and again, doubling the number of cells each time. Food provides the energy and materials, but the new cells come from the division of existing cells, each receiving the same instructions (chromosomes).

LESSON 8 (80 minutes (2 x 40-minute periods)): Why Cell Division Matters: From Baby to Teenager, and Bananas from a Test Tube

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	By the end of the lesson, the learner should be able to explain the significance of cell division in organisms (growth, repair, reproduction and development) and describe how tissue culture uses mitosis to multiply plants such as bananas and tea in Kenya.
Knowledge	Learners recognise that mitosis produces genetically identical cells that allow growth, replace worn-out or damaged cells, and support asexual reproduction; meiosis produces gametes for sexual reproduction and variation; tissue culture uses repeated mitosis of plant cells in sterile nutrient medium to produce many identical plantlets (clones); this links to how a 3 kg newborn becomes a 50 kg teenager.
Skills	Developing and using models (drawing and annotating a model linking cell number to body growth); Analysing and interpreting data (estimating cell numbers and comparing banana suckers with tissue-cultured plantlets); Constructing explanations; Engaging in argument from evidence during the discussion diamond; Communicating information through the jigsaw and gallery walk.
Attitudes	Core competencies: Creativity and Imagination (innovatively designing a tissue culture procedure and model); Digital Literacy (viewing a tissue culture or mitosis animation where available). Values: Unity (collaborating in jigsaw expert groups); Respect (appreciating others' ideas during the gallery walk). PCIs: Safety and Security (sterile technique, careful handling of hot or sharp items, handwashing); Communicable Diseases (hygiene and clean technique prevent the spread of microbes, in the tissue culture context and at home).
Key Inquiry Question	How is cell division important?
Purpose in Storyline	Lessons 1-7 gathered evidence on chromosomes, the cell cycle, mitosis and meiosis. Lesson 8 steps back to answer 'so what?': why organisms need cell division. Learners connect the mechanism of mitosis to the anchoring phenomenon (a 3 kg baby becoming a 50 kg teenager) and to a Kenyan application, tissue culture of bananas and tea. This provides the 'why' that pushes the driving question forward before Lessons 9-10 address budding, binary fission and final model revision.
Safety Notes	Wash hands before and after any activity involving plant material or media. If a demonstration with a sterile technique is done, the teacher alone handles any flame, spirit lamp or hot water; learners observe from a safe distance. No tasting of media or plant material in class. Use sealed, labelled containers if a demonstration culture is shown. Keep bags and walkways clear in a large class and carry out the gallery walk in a single direction to prevent crowding.

B. LESSON OVERVIEW

Learners begin by predicting how many times cells must divide to take a baby from 3 kg to 50 kg, and which body processes depend on cell division. They gather evidence in a jigsaw: four expert groups study growth, repair, reproduction/development, and tissue culture (with Kenyan banana and tea cases from KALRO and Kenyan farmer groups). Learners then use a discussion diamond to rank the importance of cell division, and in a gallery walk they critique each other's posters. A Driving Question Board update and a model revision show that new cells come from existing cells via mitosis, and that the same principle is exploited in tissue culture. Large-class management: work in groups of 5-6 with assigned roles, reuse printed or board-written cards, and use cold-calls on group reporters.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Intro to cell division Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall their baby photograph vs. their present look and the 14-year time-lapse. In a think-pair-share they answer on paper: (1) A baby has about 3 kg of body mass and a teenager 50 kg. What must have happened to the number of cells? (2) Name three things in your body that need new cells every day (e.g. skin after a fall on the football pitch, blood, hair). (3) A farmer in Meru wants 1,000 identical disease-free banana plants quickly. How might she get them from one plant? They sketch a quick prediction diagram of one cell becoming many.	Display the baby photo vs. teen photo or draw them on the board. Say: 'We eat ugali and sukuma wiki, yet our bodies build billions of new cells. Where do they come from?' Give individual write time of 2 minutes, then say: 'Turn to your neighbour and share.' Give WAIT TIME of 10-15 seconds after each question before calling on anyone. Cold-call 5 pairs (not volunteers) and record predictions on the board under 'Our predictions'. Do not correct yet; say: 'Hold on to this; we will test it with evidence.' In a large class, have pairs seated in rows share across the aisle, and sample 5 pairs from different parts of the room.	Think-Pair-Share and Predict-Observe-Explain (Predict step): Individual thinking exposes prior ideas (e.g. that cells just 'get bigger' or 'come from food'), and pair talk lets learners rehearse reasoning in a low-risk setting. Recording predictions creates a baseline to compare with later evidence.	Oral questions and observation: Listen for whether learners say cells increase in number versus just in size, and whether they name growth, repair and reproduction. Indicator: at least 2 of 3 uses of cell division mentioned in the pair talk. Note common misconceptions (cells come from food; growth is only cells enlarging) and tally them on the board.
Observe Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Intro to cell division Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Jigsaw. Home groups of 4-6 split into four expert groups, each with an evidence card: A) Growth: data on a human growing from a single fertilised egg (about 1 cell) to roughly 30-40 trillion cells, with an estimate of divisions (doubling: 2, 4, 8, 16...) and a photo of the child's growth. B) Repair and replacement: a case of a footballer in Kakamega healing a cut, plus facts on red blood cells and skin cell replacement; learners look at a prepared drawing/micrograph of mitosis in an onion root tip. C) Reproduction and development: budding in yeast (used in mandazi and traditional brew fermentation), binary fission, vegetative propagation of sweet potatoes and sugarcane, and meiosis making gametes, plus embryo development from zygote. D) Tissue culture: a flowchart of explant to callus to shoots to roots to hardening off, with a Kenyan	Distribute evidence cards (one set per expert group; in a large class, make 2 copies of each card and run 8 expert groups). Say: 'Your job is to become the class expert on your card. You will teach your home group in 10 minutes, so make sure everyone can explain it without reading.' Give 10 minutes in expert groups, circulate, and ask: 'Is it mitosis or meiosis here? What is your evidence?' Pause WAIT TIME of 10-15 seconds before probing. Tell home groups: 'Each expert has 2 minutes; listeners, ask at least one question.' Cold-call 4 listeners (one per card topic) to summarise another person's topic, not their own. If a demonstration of tissue culture is possible (e.g. a pre-prepared culture jar or photos from a KALRO lab), show it from the front only. Remind learners of hygiene: 'Wash hands after handling plant material.'	Jigsaw: Every learner holds a unique piece of evidence and must teach it, building accountability and shared understanding of four functions of cell division. Interdependence supports the value of Unity. It works in large classes because the teacher monitors fewer groups of experts first and then home groups.	Observation checklist and oral questions: Indicators: experts correctly link each function to mitosis or meiosis; experts use the words 'identical cells' or 'genetic information copied'; listeners' grids have four completed rows. Quick check: ask 4 cold-called learners 'Why does a tissue-cultured banana plantlet match its parent?' Expected: because cells divided by mitosis, producing genetically identical cells.

	case of tissue-cultured bananas (disease-free, uniform, faster than suckers, supplied to farmers in Kisii, Meru and Kiambu) and tea clones from Kericho. Each expert group completes a fact sheet: What is the process? Which cells divide? Mitosis or meiosis? What would happen without it? Then experts return to home groups and teach, while listeners complete a four-row summary grid.			
Explain Phase  VIDEO: Resources and population growth Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Intro to cell division Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners use a discussion diamond. Each group receives nine statements about cell division (e.g. 'Without mitosis a cut on the knee could never heal', 'Tissue culture lets Kenya produce clean banana seedlings quickly', 'Meiosis is more important than mitosis for growth', 'Cells in a baby just get bigger', 'Tissue culture plants are all identical, so one disease could affect them all', 'Binary fission is how bacteria reproduce', 'Cell division helps a seed grow into a maize plant', 'Yeast budding makes dough rise by producing more yeast cells', 'Meiosis gives variation'). They rank them in a diamond from most to least important or accurate, with a written justification for the top and bottom card. They then write a 3-sentence explanation answering: 'How does the body make the huge number of new cells needed from a 3 kg baby to a 50 kg teenager?', using evidence from the jigsaw.	Hand out statement cards (or write them on the board for groups to copy if printing is limited). Say: 'Rank these from most to least important, and be ready to defend your choice with evidence from your expert card.' Allow 10 minutes. Circulate and press: 'What evidence supports that? Could someone disagree?' After ranking, give WAIT TIME of 10-15 seconds, then cold-call 5 groups to explain their top and bottom cards. Highlight disagreement: 'Group 3 and Group 7 ranked this differently. Group 7, respond to Group 3.' Correct misconceptions: cell growth also involves division, not just enlargement; cells arise from pre-existing cells, not from food directly. Food supplies the raw materials (amino acids, glucose, minerals) and energy used to build new cells. Close by asking learners to write their 3-sentence explanation individually (3 minutes).	Discussion Diamond and Predict-Observe-Explain (Explain step): Ranking forces learners to weigh evidence and justify claims, which exposes reasoning. Comparing diamonds shows that different functions (growth, repair, reproduction, tissue culture) all rest on one mechanism: cell division. Learners return to their original prediction and revise it.	Exit ticket and oral questions: Exit ticket (collected at end of Explain): 'Name two ways cell division is important to a living organism and explain how mitosis helps tissue culture produce banana plantlets.' Success criteria: names at least two functions; states that mitosis makes identical cells; links tissue culture to repeated mitosis in nutrient medium. Observable indicators in discussion: learners cite evidence ('The card said...'), and revise their original predictions.
Driving Question Board (DQB) Creation  VIDEO: Exponential vs.	Learners return to the Driving Question Board with the question 'How does the body produce the enormous number of new cells needed for growth?' They add sticky notes or written slips in two	Say: 'Look at our DQB. Which questions can we now answer, and what new questions has today raised?' Give WAIT TIME of 10-15 seconds before taking responses. Ask each pair to write one 'now we	Think-Pair-Share linked to the DQB: Pairs consolidate their understanding and generate questions, giving learners ownership of the inquiry. Sorting questions into 'answered' and	Oral questions and DQB review: Indicators: slips include correct statements such as 'mitosis produces identical cells' and new questions that move beyond recall (cell cycle control, cancer,

linear growth over time Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Intro to cell division Source: Khan Academy (English - US curriculum) Search ARES for similar readings	columns: 'What we now know' (e.g. 'New cells come from existing cells dividing by mitosis'; 'Cell division is needed for growth, repair, and asexual reproduction'; 'Tissue culture makes identical plantlets by repeated mitosis') and 'New questions' (e.g. 'How do cells know when to stop dividing?'; 'Why do some cancers make cells divide uncontrollably?'; 'How do budding in yeast and binary fission differ from mitosis?'; 'Can tissue-cultured crops lose variety and become vulnerable?'). Pairs choose one old question to mark as answered.	know' and one 'new question'. Cold-call 4 pairs to place and read their slip aloud and ask the class: 'Do we agree? What evidence supports it?' Group the slips by theme on the board (growth control, cancer, microbes, genetic variety). Flag the questions that Lesson 9 (budding and binary fission) and Lesson 10 (final model) will address. In a large class, use one representative per row to post slips on behalf of the group.	'new' makes progress visible and shows how today's evidence advances the driving question.	variation). Record how many pairs wrote a testable or deeper question. Flag any slip showing the misconception that cells arise from food.
Model Building Phase VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Intro to cell division Source: Khan Academy (English - US curriculum) Search ARES for similar readings	In groups, learners revise their cumulative model of how a single cell becomes a whole body, adding today's evidence. The model is a labelled poster/diagram with: (1) a fertilised egg (zygote) dividing repeatedly by mitosis into many cells (use 2, 4, 8, 16 ...); (2) arrows showing growth, repair (a healing cut) and replacement of cells; (3) differentiation into specialised cells; (4) meiosis producing gametes for reproduction; (5) a side panel showing tissue culture: explant to callus to plantlets to hardened banana or tea plants in Kenyan farms. Groups write one caption: 'Food supplies ____, and cell division supplies ____.' Gallery walk: groups post posters, rotate in one direction, and leave two sticky-note comments (one 'star' and one 'wonder') on each poster they visit, showing respect for others' work.	Say: 'Revise your model so that someone from another school could use it to explain how a 3 kg baby becomes a 50 kg teenager, and how a farmer in Meru can get thousands of banana plantlets.' Provide manila paper or exercise-book pages for large classes; if materials are limited, allow one poster per group of 6. Allow 12 minutes to build. Run the gallery walk for 10 minutes with a signal to rotate every 2 minutes; instruct: 'Move clockwise; write one star and one wonder.' Run a model check while circulating: ask 'Where in your model do new cells come from? What does mitosis do to the chromosomes?' Use WAIT TIME of 10-15 seconds, and cold-call 3 groups to explain a revision they made after reading others' comments. Close with an exit ticket: 'Complete: Cell division is important because _____. Tissue culture works because _____. 'Remind learners to wash hands and keep walkways clear.	Gallery Walk and model revision: Viewing and critiquing other groups' models exposes alternative representations and surfaces gaps (e.g. missing meiosis or the role of food). Revising the model after peer feedback makes learning cumulative, ties the mechanism of mitosis to real Kenyan applications, and models the Respect value through constructive feedback.	Model checks and exit ticket: Indicators of a strong model: shows cell number increasing through repeated mitosis; labels growth, repair and reproduction; indicates daughter cells are genetically identical; includes the tissue culture steps in correct order; includes food as the source of raw materials. Exit ticket success: learners accurately finish both sentence stems. Sort tickets into 'secure', 'nearly there' and 'needs reteaching' to plan the start of Lesson 9.

D. TEACHER REFLECTION

After the lesson, consider: (1) Did learners move from 'cells get bigger' to 'cells increase in number by mitosis'? Which learners still hold the misconception? (2) Did the jigsaw work in the large class, and were all expert groups able to access evidence cards? What adjustments (extra copies, board summaries) are needed? (3) Were learners able to connect tissue culture to repeated mitosis and identical clones, and did anyone raise the risk of reduced genetic variety? (4) Which DQB questions are ready to be explored in Lesson 9 (budding and binary fission) and Lesson 10? (5) Was WAIT TIME of 10-15 seconds used consistently, and were cold-calls spread across the room? (6) How well were safety and hygiene reinforced?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that growth, repair of injured tissues, and reproduction in organisms such as yeast and plants all involve an increase in the number of cells. We saw how a tissue culture of a banana or tea plant starts with a small piece of plant (explant) that grows into a callus, then shoots and roots, forming many identical plantlets.
What did I learn?	We learned that cell division by mitosis produces genetically identical cells for growth, repair and asexual reproduction, while meiosis produces gametes for sexual reproduction and variation. Tissue culture uses repeated mitosis in sterile nutrient medium to produce many disease-free clones quickly. Food provides the raw materials and energy; cell division provides the new cells.
How does this explain the phenomenon?	The body produces the enormous number of new cells needed for growth because existing cells divide repeatedly by mitosis, starting from one fertilised egg, doubling the cell number again and again. Each new cell receives a full copy of the chromosomes, so it has the right instructions. The same principle lets farmers in Kenya multiply bananas and tea through tissue culture.

LESSON 9 (80 minutes (2 x 40-minute periods)): Consolidation and Digital Animations: Comparing Mitosis and Meiosis to Explain Growth

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To consolidate evidence from Lessons 1-8 by watching animations of mitosis and meiosis, comparing the two processes in a table, and presenting chromosome-behaviour models to peers with respectful feedback, so learners can explain how the body produces billions of new cells for growth.
Knowledge	a) Explain the significance of mitosis and meiosis. b) Illustrate the stages of mitosis and meiosis in cells. d) Recognise the importance of cell division in reproduction, growth, and development.
Skills	Developing and using models (NGSS SEP 2); Analysing and interpreting data from animations (SEP 4); Constructing explanations (SEP 6); Engaging in argument from evidence and giving peer feedback (SEP 7); Obtaining, evaluating and communicating information (SEP 8).
Attitudes	Core competency, Digital Literacy: the learner enhances computing skills while watching animations showing various stages of mitosis and meiosis. Value, Respect: the learner appreciates others' efforts while presenting illustrations to show the behaviour of chromosomes in various stages of cell division. Value, Unity: the learner collaborates with others in groups.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	By Lesson 9 learners have gathered evidence about chromosomes, mitosis, meiosis, budding, binary fission and tissue culture. This lesson brings it together: animations let learners see in motion what they modelled and observed under the microscope, the comparison table shows which process explains growth (mitosis) and which explains gamete formation (meiosis), and the presentations let learners test their models against peers' evidence. It advances the driving question 'How does the body produce the enormous number of new cells needed for growth?' by letting learners explain how one fertilised cell becomes billions of identical cells that each carry the right instructions, and prepares for the final consolidated model and explanation in Lesson 10.
Safety Notes	Safety and Security PCI: Keep bags and cables away from walkways when using a projector, shared tablets or phones. Handle any devices with dry hands and do not overcrowd the power sockets. If using the school computer lab, follow lab rules. In the gallery walk, move slowly and keep aisles clear. Use blunt-tipped scissors and non-toxic coloured pencils or chalk for models. Learners wash hands after handling any pipe cleaners, wool or soil-contaminated materials from earlier lessons.

B. LESSON OVERVIEW

Anchoring phenomenon: a Kenyan child grows from a 3 kg newborn into a 50 kg teenager, requiring billions of new cells, yet we only eat food such as ugali, sukuma wiki and beans. In this lesson learners predict what animations of mitosis and meiosis will show about how cells multiply, observe the animations (a single projected animation for the whole class, or one device per group of 6-8 if the class is large), complete a mitosis vs meiosis comparison table through a jigsaw, present chromosome-behaviour models in a gallery walk with respectful feedback, update the Driving Question Board, and revise their class model of how one cell becomes billions. Large-class strategies: groups of 6-8, expert groups, one gallery per half of the room, and a single shared animation projected so no learner depends on personal devices.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners recall the baby photograph and time-lapse of a child growing up over 14 years. In a Think-Pair-Share they predict: 'If we could film inside a growing child's body, what would we see happening to one cell, and what would happen to the chromosomes inside it?' Each learner sketches a quick 3-box prediction (before, during, after one cell divides) in the notebook, showing how many chromosomes are in each new cell. Then they predict one difference between how a cell divides to make skin for a growing child and how it divides to make gametes (sperm and egg).	Display the baby photograph and say: 'Remember the newborn of 3 kg who becomes a teenager of 50 kg. Think silently: what is happening to one cell inside Wanjiru's growing leg bone?' Give WAIT TIME of 10-15 seconds before any talk. Say: 'Turn to your neighbour and share your sketch for 2 minutes.' Circulate and listen for misconceptions such as 'cells just get bigger' or 'the chromosomes get halved in every division'. Cold-call 4 pairs from different parts of the room: 'Amina and Otieno, what did you predict happens to the chromosomes?' Record predictions on the board under 'Our Predictions' without judging them. Say: 'We will check these predictions against the animations.'	Think-Pair-Share combined with Predict-Observe-Explain (the Predict step). Individual thinking first gives every learner in a large class a voice, pairing rehearses ideas safely, and recording predictions creates something to test against evidence, so learners see the animations as a way to answer their own questions about the growth phenomenon.	Oral questions and observation. Look for: (a) learners who sketch the same chromosome number in parent and daughter cells for growth, (b) learners who show chromosomes copying before division, (c) learners who say cells get bigger only. Tally on a class list the number of learners holding each idea (e.g., ticks for 'same number', 'half number', 'cells just enlarge') to plan Explain Phase questions.
Observe Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Mitosis vs. Meiosis (Flexbook) Source: CK-12  Search ARES for similar readings	Learners watch animations of mitosis and then meiosis (projected once for the class, replayed in stages; or on shared devices in groups of 6-8 where available, building Digital Literacy). While watching, each learner completes a stage-tracking sheet noting: number of chromosomes at the start, what happens to the chromosomes at prophase, metaphase, anaphase and telophase, and number and type of cells at the end. Then, in a Jigsaw activity, the class forms home groups of 4, and each member joins an expert group: Expert A, Mitosis stages; Expert B, Meiosis I; Expert C, Meiosis II; Expert D, Significance (growth, repair, gametes, variation). Experts compare observations, agree on key points and then return to teach	Say: 'Watch the animation for the first time just to see the whole process. Do not write yet.' Play once without pausing. Then say: 'Now we play it again and pause at each stage. Write what you see happening to the chromosomes.' Pause at each stage and ask: 'How many chromosomes can you count now?' WAIT TIME of 10-15 seconds, then cold-call 3 learners per stage. For jigsaw, say: 'Number yourselves 1 to 4 in your home group. Number 1s go to the front-left corner, 2s to the front-right, 3s to the back-left, 4s to the back-right.' Give 8 minutes for expert work and 10 minutes for teaching back, with each expert speaking for 2 minutes. Circulate and ask: 'What evidence in the animation tells you that?' If devices are scarce, replay the animation	Jigsaw with a comparison table. Each learner becomes responsible for part of the evidence, so all must contribute, which is vital for large classes. Comparing the two processes in one table helps learners notice that mitosis keeps the chromosome number the same (explaining growth of the child from a 3 kg baby) while meiosis halves it (explaining why gametes can combine at fertilisation without doubling chromosome number each generation).	Model checks and oral questions. Observable indicators: stage-tracking sheet correctly shows 2n to two 2n cells in mitosis and 2n to four n cells in meiosis; table entries correctly state mitosis = 1 division, 2 identical diploid cells; meiosis = 2 divisions, 4 genetically different haploid cells. Experts who can explain chromosomes lining up at the equator in metaphase and sister chromatids separating in anaphase without reading from notes demonstrate understanding. Note groups placing growth under meiosis for follow-up.

	their home group. Home groups complete a comparison table of mitosis and meiosis: where it occurs (e.g., root tip of a maize seedling, skin, bone marrow vs testes in a bull or ovaries in a cow), number of divisions, number of daughter cells, chromosome number (diploid 2n or haploid n), crossing over, genetic identity, and role (growth, repair, gamete formation).	for each stage as many times as requested. Provide a pre-drawn blank comparison table on the board and on manila paper. Select 4 home groups at random to share one row of the table each, and ask the class: 'Do you agree? Show thumbs up, sideways or down.'		
Explain Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mitosis vs. Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners return to their predictions and compare them with the animation evidence. In a Discussion Diamond, each group of 4 ranks nine statement cards in a diamond from 'most useful for explaining the child's growth' to 'least useful'. Example cards: 'Chromosomes are copied before mitosis', 'Mitosis makes two identical cells', 'Meiosis halves the chromosome number', 'Cells get bigger only', 'Each new cell receives a full set of instructions', 'Crossing over mixes genes', 'Food provides materials and energy for making new cells', 'Meiosis makes the cells that make the skin', 'Mitosis happens in the growth zone of bones and in the root tips of maize'. Groups justify their top and bottom choices using evidence from the animations. Learners then write a 4-sentence explanation: 'How does one fertilised egg become the billions of cells in a 50 kg teenager, and how does each cell get the right instructions?'	Hand out the nine cards in envelopes (one set per group of 4, or per group of 6-8 in a large class, with a second set of cards on the board). Say: 'Arrange your cards in a diamond: one at the top for the most useful idea, two in the next row, three in the middle, two below, and one at the bottom. You must agree as a group and be ready to defend your choices with evidence.' Give 10 minutes. Circulate and ask: 'Why did you place that card there? Which part of the animation supports it?' Then say: 'Now return to your predictions on the board. Which ones need changing?' WAIT TIME of 10-15 seconds. Cold-call 4 groups to explain their top card and 2 groups to explain their bottom card. Ask: 'Some of you said food gives us new cells. What role does the food really play?' Guide learners to say that food (ugali, beans, sukuma wiki) supplies energy and raw materials, but the cells themselves come from existing cells by mitosis. Ask learners to write their 4-sentence explanation individually in 5 minutes.	Discussion Diamond plus Predict-Observe-Explain (the Explain step). Ranking forces learners to weigh the relative importance of ideas and justify choices with evidence, surfacing misconceptions in group talk. Returning to initial predictions makes the shift in thinking visible and supports building a causal explanation of growth.	Oral questions and written explanations. Observable indicators: learners state that new cells come from existing cells by mitosis; that DNA is copied first so each daughter cell gets a complete, identical set of chromosomes (instructions); that food supplies energy and raw materials; that meiosis is for gametes, not growth. Look for learners who connect the number of mitosis rounds to cell numbers (1, 2, 4, 8, 16... doubling). Record the number of learners using evidence words such as 'the animation showed...'.
Driving Question	Learners return to the Driving Question Board. Each group reads	Say: 'Go back to our Driving Question Board. In your groups	Driving Question Board sorting. Sorting questions as answered,	Exit tickets. Success indicators: Prompt 1 correctly names at least

Board (DQB) Creation  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	the questions posted since Lesson 1 (for example, 'How do cells know when to stop dividing?' and 'Why do cuts on the knee heal?'). Groups move sticky notes or paper slips into three columns: 'Answered', 'Partly answered', and 'New questions'. They add new questions raised by the animations, such as 'What happens if mitosis goes wrong, as in cancer?' or 'Why do identical twins from one fertilised egg look alike while siblings do not?' Each learner completes a short exit ticket on a slip of paper.	decide which questions we can now answer using evidence from today.' Give 6 minutes. Send one representative per group to place slips on the board (to limit crowding in a large class). Cold-call 3 groups: 'Read one question you moved to Answered, and tell us the evidence.' WAIT TIME of 10-15 seconds before calling. Then say: 'Write one new question on a slip and put it in the box on your way out.' Distribute exit tickets with three prompts: (1) One difference between mitosis and meiosis. (2) Which process helps a child grow, and why? (3) One question I still have.	partly answered or new shows learners how their understanding grows and keeps the anchoring phenomenon of the child's growth as the purpose of the investigation. Learner questions guide the next lesson.	one difference (e.g., mitosis gives 2 identical diploid cells; meiosis gives 4 different haploid cells); Prompt 2 names mitosis and links it to growth and repair; Prompt 3 shows curiosity linked to the phenomenon. Sort tickets into three piles (secure, developing, needs support) and use them to plan groups for Lesson 10.
Model Building Phase  VIDEO: Comparing mitosis and meiosis Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Groups present their chromosome-behaviour models (made in earlier lessons from wool, pipe cleaners, bottle tops, drawings on manila paper, or charts showing mitosis and meiosis stages) in a Gallery Walk. Half of each group stays beside the model as presenter for 5 minutes, while the others tour and then swap. Visitors give feedback using the structure 'Two stars and a wish': two things done well and one suggestion, and apply the value of Respect by appreciating others' efforts. Groups then revise their model using feedback. Finally, the whole class updates the class model, 'How one cell becomes billions', by adding: chromosome copying before division, equal sharing in mitosis, repeated rounds of mitosis to give billions of cells, and meiosis in the formation of gametes before fertilisation. Learners add a label showing where in the body each process happens.	Say: 'Place your models and drawings on your desks or on the wall. Half of you stay to explain, half go and visit. We swap after 5 minutes.' Divide the room into two galleries (A and B) to reduce congestion in a large class. Model respectful feedback: 'I liked how your anaphase shows the chromatids moving to opposite poles. One suggestion is to label the spindle fibres.' Post sentence starters: 'I appreciate...', 'I wonder...', 'Could you show...'. Circulate with a checklist: chromosome number correct at each stage, chromosomes copied before division, sister chromatids separating, correct number of final cells, labels present. Ask: 'Where on your model can I see why a child can grow from one cell?' WAIT TIME of 10-15 seconds. Cold-call 3 learners to share a piece of feedback they received and how they will use it. Then lead the class model revision: draw a flow diagram on the board with learner contributions and say:	Gallery Walk with peer feedback and cumulative model revision. Presenting and critiquing models lets learners test their reasoning against others' ideas, correct errors in chromosome behaviour and build a shared class model that explains how growth is possible through mitosis while meiosis supports reproduction. The respectful feedback routine nurtures the Respect value.	Model checks. Observable indicators: model shows DNA/chromosome replication before division; chromosomes line up at the equator; sister chromatids pull apart in anaphase; mitosis produces 2 identical diploid cells; meiosis produces 4 haploid cells with crossing over; labels used correctly (centromere, spindle, chromatid). Feedback is specific, kind and evidence-based. Groups revise at least one element of their model after feedback. Record which groups need extra support for Lesson 10.

		'Which part of our explanation of the child's growth does this piece of the model support?' Close by saying: 'In Lesson 10 we will use this model to write a complete explanation of the phenomenon.'		
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D. TEACHER REFLECTION

1) Which prediction ideas (for example 'cells only get bigger', or 'chromosomes halve in every division') persisted after the animations, and which learners held them? 2) Did the jigsaw ensure that every learner, including in a large class, contributed to the comparison table? 3) Could learners use evidence from the animation to explain why a Kenyan child can grow from 3 kg to 50 kg, and why food alone does not produce cells? 4) How well did the gallery walk and feedback show Respect, and how did feedback change the models? 5) What should be revisited in Lesson 10 (for example, the role of the cell cycle, errors in division, tissue culture), based on exit ticket results? 6) What adjustments to grouping, device sharing or time are needed?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We watched animations of mitosis and meiosis and saw chromosomes copy, line up, separate and move into new cells. In mitosis one cell made two identical cells with the same number of chromosomes; in meiosis one cell made four cells, each with half the number of chromosomes. We saw models of these stages from our classmates.
What did I learn?	Mitosis produces genetically identical diploid cells for growth and repair, such as bone, skin and the root tips of maize. Meiosis produces haploid gametes with different gene combinations for sexual reproduction. DNA is copied before division so that each new cell receives a full set of instructions. Food gives energy and raw materials but does not turn into cells by itself.
How does this explain the phenomenon?	A child grows from 3 kg to 50 kg because the fertilised egg divides again and again by mitosis, doubling cell numbers each round (1, 2, 4, 8, 16...) until billions of cells exist. Each new cell receives an exact copy of the chromosomes, so it has the right instructions to work properly. Meiosis earlier formed the egg and sperm that joined to make the first cell, keeping the chromosome number constant from one generation to the next.

LESSON 10 (80 minutes (double lesson: 2 x 40 minutes)): Final Model and Return to the Phenomenon: How Does the Body Make the Billions of New Cells Needed for Growth?

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners complete and present their final cumulative model of cell division, compare it with their Lesson 1 drawing, close out the Driving Question Board, and write an evidence-based explanation of how a 3 kg newborn becomes a 50 kg teenager.
Knowledge	a) explain the significance of mitosis and meiosis; b) illustrate the stages of mitosis and meiosis in cells; c) distinguish between budding and binary fission in relation to cell division; d) recognise the importance of cell division in reproduction, growth, and development.
Skills	Developing and using models (revising and presenting the final model); constructing explanations using evidence (Claim-Evidence-Reasoning); engaging in argument from evidence during the gallery walk; obtaining, evaluating and communicating information (jigsaw). Core competencies: Creativity and Imagination (innovatively modelling budding and binary fission); Digital Literacy (using what was learned from the mitosis and meiosis animations).
Attitudes	Unity: collaborating with peers while building, defending and critiquing models. Respect: appreciating others' efforts while presenting illustrations of chromosome behaviour in the stages of cell division. Curiosity and willingness to revise ideas in the light of evidence.
Key Inquiry Question	How is cell division important?
Purpose in Storyline	This is the culminating lesson. Learners return to the anchoring phenomenon, a child growing from a 3 kg newborn to a 50 kg teenager, and use all the evidence from Lessons 1-9 (onion root tip and microscope observations, chromosome models, mitosis and meiosis animations, yeast budding, bacterial binary fission, tissue culture) to answer the Driving Question: 'How does the body produce the enormous number of new cells needed for growth?' They compare Lesson 1 and final models to see how their thinking has changed.
Safety Notes	Safety and Security: if microscopes are used for the model check station, carry them with two hands, keep slides away from table edges and never point the mirror or lamp at the sun. Wash hands with soap and clean water after handling any yeast or culture materials from earlier lessons, and do not taste any materials (Communicable Diseases PCI). Keep bags and stools out of walkways during the gallery walk so that large classes can move safely.

B. LESSON OVERVIEW

In this final double lesson, learners return to the baby photographs and growth time-lapse from Lesson 1. They first predict, individually and then in think-pair-share, how a 3 kg baby could become a 50 kg teenager, using only the ideas they hold now. In the Observe phase, learners use a jigsaw to re-examine the key evidence: expert groups on (1) mitosis and the cell cycle, (2) meiosis and the zygote, (3) budding and binary fission, and (4) tissue culture. They then teach their home groups. In the Explain phase, learners use a discussion diamond to rank the most important ideas for explaining growth and write a Claim-Evidence-Reasoning Final Explanation. The class then closes the Driving Question Board, matching each question with the lesson that answered it and noting new questions. Finally, groups finalise their cumulative model, display it for a gallery walk, and compare it with their Lesson 1 drawing. The class is large, so all work is in groups of 5-6 using shared materials, with group roles (recorder, reporter, materials manager, timekeeper, checker) and a maximum of 8 groups.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Zygote differentiating into somatic and germ cells Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Learners look again at their baby photograph (or the class photo display prepared earlier) next to the growth time-lapse still, and at their own Lesson 1 drawing of how a baby becomes a teenager. Individually, they take 3 minutes to write a short prediction on a sticky note or in the exercise book: 'If you could zoom into the body of a growing child, what would be happening inside to make the body grow from 3 kg to 50 kg? What do you think happens in one cell to make two cells?' They then do a think-pair-share with a partner and then join their group of 5-6 to agree on one 'best prediction' to be written on manila paper. They also hold a short debate on a common idea such as 'Cells just get bigger as we eat ugali, githeri and sukuma wiki'. This connects the phenomenon back to the puzzle: we only eat food, yet our bodies make billions of new cells.	Display the baby photographs and the time-lapse still. Say: 'In Lesson 1 you told me that a baby of about 3 kg becomes a teenager of 50 kg. Today you will finally answer: where do all the new cells come from?' Give the individual writing task: 'Write, in 3 minutes, what you think happens inside the body.' WAIT TIME 10-15 seconds after asking: 'Do cells just grow bigger, or does their number increase? What evidence do you have?' Use the think-pair-share: pairs talk for 2 minutes. Cold-call 4 pairs (from different parts of the class so that quiet learners are included): 'Pair 3, what did your partner say that made you think differently?' Record ideas on the board under two columns, 'cells get bigger' and 'cells increase in number', without judging them. Circulate and listen for misconceptions such as 'food turns directly into new cells' or 'a muscle cell divides to make a muscle cell, with half the chromosomes'. Say: 'Keep your prediction; at the end of the lesson you will check it against all our evidence.'	Think-Pair-Share combined with Predict-Observe-Explain (POE). Writing individually first ensures that every learner commits to an idea before hearing others, which is essential in a large class where a few voices usually dominate. Pair talk and then group consensus let learners test ideas on a smaller audience and expose competing explanations (growth by cell enlargement versus growth by cell number). The prediction is the baseline against which learners will later judge how much their thinking has changed.	Oral questions and a quick scan of written predictions. Look for: (a) learners who mention 'cells divide' or 'cells multiply' without specifying how, (b) learners who say that cells only get bigger, (c) use of terms such as mitosis, chromosome, DNA or copy. Tally on a class list (tick or cross) how many predictions mention an increase in cell number, to compare with the end of the lesson. Indicator: at least half of the groups can state that new cells come from existing cells.
Observe Phase  VIDEO: Chromosomes, chromatids and chromatin Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: The cell cycle and mitosis	Learners carry out a jigsaw activity to review the evidence collected in Lessons 1-9. Home groups of 4-6 are divided into four expert topics: (1) Mitosis and the cell cycle (onion root tip slides or photomicrographs, chromosome models, the sequence prophase, metaphase, anaphase, telophase, cytokinesis; the production of two identical daughter cells with the same chromosome number), (2) Meiosis and fertilisation (the formation of gametes with half the	Set up the room before learners arrive: label four expert stations with the evidence packs (charts, onion root tip photomicrographs, chromosome models made from wool/pipe cleaners, tissue culture photographs, yeast and bacteria diagrams). Because the class is large, assign 2 expert groups to each topic if there are more than 4 groups, and a maximum of 6 per group. Say: 'You are now the experts. Your job is to teach your home group how your evidence	Jigsaw. Each learner becomes responsible for one piece of the evidence and must teach it, which builds accountability and ensures that all pieces of the puzzle (mitosis, meiosis, budding/binary fission, tissue culture) are assembled into a single coherent explanation. The structure also manages a large class, because many learners talk at the same time in small groups rather than waiting to hear from the teacher. Predict-Observe-Explain: learners	Oral questions and observation checklist. Indicators: expert learners (a) correctly sequence the stages of mitosis, (b) state that mitosis gives two identical daughter cells with the same chromosome number, (c) state that meiosis gives four cells with half the chromosome number, (d) distinguish budding (unequal division with a bud) from binary fission (equal splitting into two), and (e) link tissue culture to the idea that a single cell can divide to

Source: Khan Academy (English - US curriculum) Search ARES for similar readings	chromosome number, fusion to form the zygote, and the 46-chromosome human cell), (3) Budding in yeast (from mandazi dough or mursik/busaa fermentation) and binary fission in bacteria, and the difference between them, and (4) Tissue culture (banana, tea cuttings from Kericho or sukuma wiki seedlings grown from a single cell or piece of tissue). Expert groups spend 12 minutes using their notes, models, charts, diagrams and photomicrographs to prepare a 2-minute explanation and a labelled sketch showing 'how this evidence helps explain the child's growth'. Learners then return to the home groups and teach their topic in 3 minutes each, while the others take notes in a 'What we observed / What it tells us' table. Learners should confirm key evidence: a zygote (one cell) divides repeatedly, so 1 becomes 2, 4, 8, 16... and after about 40+ rounds there would be trillions of cells.	explains the 50 kg teenager. You have 12 minutes.' Circulate and ask each expert group: 'How many cells do you start with, and how many do you end with? How do you know the daughter cells are identical?' WAIT TIME 10-15 seconds after each question. Ask the group to use the evidence rather than recall. After the expert phase, give the signal to return to home groups and say: 'Teach, do not read. Use your diagram.' Cold-call 3 learners at random: 'Tell me one thing you learned from another group's expert that you did not know before.' Prompt connections: 'How does a doubling pattern (1, 2, 4, 8, 16...) help explain 3 kg to 50 kg?' Use a countdown timer visible to all. If microscopes are used, limit to 1 per group and enforce the safety rules.	compare the evidence with their earlier predictions.	form a whole organism. Tick on the class list those who teach accurately. Any group that confuses mitosis and meiosis is revisited with a chromosome model during circulation.
Explain Phase  VIDEO: Asexual and sexual reproduction Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12 Search ARES for similar readings	Learners use a discussion diamond to decide which ideas are the most important for explaining how the body produces billions of new cells. Each group receives 9 cards with ideas (for example: DNA is copied before division; mitosis produces two identical cells; chromosomes line up and separate; the zygote forms from two gametes made by meiosis; cells divide repeatedly so numbers double; cells differentiate into muscle, bone and blood cells; food (ugali, beans, sukuma wiki, milk) provides raw materials and energy; cells also get bigger; budding and binary fission are	Distribute the card sets. Say: 'Rank these ideas as a group. You must be ready to defend why your number 1 idea matters most for explaining the growth of a child.' Remind them that there is no single correct pattern but that the reasoning must be based on evidence. Circulate and challenge: 'Why did you put food lower than DNA copying? Can cells divide without raw materials?' WAIT TIME 10-15 seconds. Cold-call 3 groups to explain their top and bottom cards, then ask other groups to respond with 'agree/disagree and why'. Name two of the discussion norms:	Discussion Diamond. Ranking forces learners to weigh the relative importance of ideas and to justify their reasoning, moving from a list of facts to a causal explanation. It also gives every member of a large group a voice, because the diamond cannot be completed without agreement. The CER writing then individualises the sensemaking so that each learner demonstrates their own understanding. Predict-Observe-Explain is completed here when learners explain why their initial prediction was correct, incomplete or wrong.	Model check on the written Final Explanation using a 4-point rubric: (1) states that new cells are produced by division of existing cells by mitosis; (2) cites at least three pieces of evidence from earlier lessons; (3) explains that DNA/chromosomes are copied so that each daughter cell receives identical instructions; (4) links repeated division, differentiation and food as raw materials to growth from 3 kg to 50 kg. Exit ticket (last 3 minutes): 'In one sentence, how does a baby's body make billions of new cells? Write one thing you are still wondering about.' Collect for review: success

	asexual reproduction in yeast and bacteria). In groups, they rank them in a diamond shape (1 most important at the top, 2 in the next row, 3 in the middle, 2 below, 1 least important at the bottom), and must justify their choices. They then write individually a Final Explanation using Claim-Evidence-Reasoning (CER) in 10 minutes: Claim: how the body produces the enormous number of new cells; Evidence: at least three pieces from earlier lessons (e.g. onion root tip cells in different stages, chromosome models, animations, yeast budding); Reasoning: link between the evidence and the science ideas (DNA replication, mitosis, identical chromosomes, growth by repeated division, food as raw materials, and why every new cell receives the right instructions). A model answer from a Kenyan case: 'A child in Kisumu eats ugali with omena and sukuma wiki. The food gives the materials and energy. The zygote divides by mitosis again and again...' Learners then check the explanation with a partner using a checklist.	'Respect each person's idea; we disagree with ideas, not people.' Then introduce the CER frame on the board: 'My claim is ... The evidence is ... This shows that ... because ...'. Give a sentence starter for support. Say: 'Your explanation must answer both parts of our phenomenon: where do the cells come from, and how does each new cell get the right instructions?' While learners write, circulate and read at least 8 explanations over the shoulder, noting common gaps (e.g. not mentioning DNA replication before mitosis). Select two anonymous examples to show on the board for a class critique. Do a brief explicit clarification of any remaining misconception, for example: 'Meiosis makes gametes, not body cells; growth is by mitosis.'		= at least 3/4 rubric points for most learners; those with 0-1 points get a follow-up conference.
Driving Question Board (DQB) Creation  VIDEO: Chromosomes, chromatids and chromatin - dubbed(KY) Source: Khan Academy (Kiswahili)  Search ARES for similar videos  EXERCISE:	Learners revisit the Driving Question Board that has been built across the unit. Each group is given a set of 3-4 sticky-note questions from the board (the questions were posted in Lesson 1 and later). They decide whether each question has been answered, partly answered or not yet answered, and move it to the correct column: 'Answered', 'Partly answered', 'New questions'. For answered questions, a group member writes the answer and the evidence (and the lesson number	Point to the DQB. Say: 'Look at the questions you asked in Lesson 1. Which can we now answer with evidence?' Give each group 3-4 questions and 5 minutes. Say: 'To move a question to Answered, you must name the evidence and the lesson.' WAIT TIME 10-15 seconds after asking: 'How do you know this question is fully answered and not partly answered?' Cold-call 4 groups to report on one question each (one from each column). Challenge a vague answer: 'Which observation	Driving Question Board review (supported by Oral Questions and Think-Pair-Share within groups). Sorting questions by answered status makes the growth of class knowledge visible and shows learners that scientific understanding develops as evidence builds. Linking each answer to a specific lesson and observation reinforces evidence-based reasoning and shows how the lessons fit together in one storyline.	Oral questions and observation of DQB sorting. Indicators: groups cite specific evidence (e.g. 'onion root tip cells in metaphase in Lesson 4') rather than only stating an answer; they correctly place questions that need more investigation in the 'partly answered' or 'new' column; at least one new, testable question per group. Note groups that move a question to 'Answered' without evidence and ask them to revisit.

Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	in which it was answered) on the note or on a new colour of sticky note. Examples of questions: 'Where do the new cells come from?', 'How does each new cell get the right instructions?', 'Why do we need food if cells divide?', 'Why do cuts on the knee heal?', 'Can plants like bananas be grown from one cell?'. Learners also add at least one new question (for example: 'What happens if cell division goes wrong, such as in cancer?', 'How do identical twins form?', 'Why do athletes like Eliud Kipchoge have more muscle repair?'). The class discusses which new questions can be investigated in future units.	supports that? Could someone from another group add more evidence?' Record the number of questions moved to Answered, and highlight at least one question the class cannot yet answer, such as cell division gone wrong in cancer, to show that science keeps raising new questions. Ensure everyone remains involved by assigning the roles of reader, writer and reporter. Validate respectful listening: 'Thank you for appreciating the efforts of your classmates.' Because the class is large, use two sections of the board or a wall so groups do not crowd.		
Model Building Phase  VIDEO: Interphase Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Cell Cycle, Mitosis, and Meiosis Practice Source: CK-12  Search ARES for similar readings	Groups finalise their cumulative model of how the body produces new cells for growth. The model is drawn on a large manila paper in a flow diagram form with labels: (1) a food input (ugali, beans, sukuma wiki, milk) providing raw materials and energy; (2) the zygote formed by fertilisation of gametes made by meiosis, each with half the number of chromosomes; (3) DNA replication in interphase; (4) the stages of mitosis (prophase, metaphase, anaphase, telophase) with chromosomes, spindle fibres and cytokinesis; (5) two identical daughter cells; (6) repeated cycles shown as 1, 2, 4, 8, 16... leading to billions of cells; (7) differentiation into specialised cells (muscle, bone, nerve, blood); (8) links to other types of division (budding in yeast, binary fission in bacteria, tissue culture in bananas) as a side panel; and (9) the child at 3 kg and at 50 kg. Groups include a key and an 'evidence box' stating which lesson gave each idea.	Distribute manila paper, markers and the Model Checklist (see below) and tell learners: 'Your model must explain both parts of our phenomenon: where the new cells come from and how each new cell gets the right instructions. If your model cannot explain it, it is not finished.' Give 12 minutes to finalise. Circulate and use probing questions: 'Show me where DNA is copied in your model. Where do the two identical daughter cells get their instructions? What would happen if chromosomes did not separate evenly?' WAIT TIME 10-15 seconds. Set up the gallery walk for a large class: split into two shifts (Group A presents while B walks, then switch), 5 minutes per shift, with one rotating presenter per group. Say: 'Visitors: give two stars and one wish, and be respectful and specific.' Cold-call at least 5 learners during the walk to explain a feature of another group's model. Debrief by projecting or holding up one	Gallery Walk with cumulative model revision. Moving among models exposes learners to alternative representations, prompts argument from evidence and encourages revision based on peer feedback. Comparing the Lesson 1 drawing with the final model makes conceptual change visible and builds metacognition: learners can say not only what they know, but how and why their thinking changed. This also covers the Creativity and Imagination competence when learners innovatively represent budding in yeast and binary fission.	Model checks using a checklist (tick each: food as raw material; zygote from gametes by meiosis; DNA replication before division; four stages of mitosis in correct order with chromosome behaviour; two identical daughter cells; repeated doubling; differentiation; budding/binary fission/tissue culture link; evidence box with lesson numbers). Success criteria: at least 7 of 9 items present in most groups. Oral questions during the gallery walk: look for learners who explain using the model rather than reading labels. Exit ticket: 'One thing I used to think, and what I think now.' Indicator of conceptual change: the final model includes DNA copying and mitosis, where the Lesson 1 drawing showed only a growing or enlarging body.

	They display the models for a gallery walk: half the group stays to explain (rotate after 5 minutes), the others visit other models with sticky notes (two 'stars' for strengths and one 'wish' for improvement) and use the checklist to compare. Back at their own models, groups read feedback and make final revisions. Finally, each learner places their Lesson 1 drawing next to the final model and writes in the exercise book three changes: 'What I first thought / What I know now / The evidence that changed my mind'. Respect is shown by appreciating the efforts of other groups when presenting the illustrations of chromosome behaviour.	Lesson 1 drawing and a final model side by side (use a volunteer's work with consent). Ask: 'What is different? What evidence changed the model?' Close by returning to the phenomenon: 'So how does a 3 kg baby become a 50 kg teenager?' Take 3 learner responses in 1 sentence each, then summarise: 'By repeated mitosis of existing cells, with DNA copied each time so that new cells have the right instructions, supported by food, and followed by differentiation.' Collect Final Explanations and exit tickets, then thank learners for collaboration (Unity).		
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D. TEACHER REFLECTION

After the lesson, reflect on: (1) How many learners moved from 'cells just get bigger' to 'cells increase in number by mitosis' (compare the tally from the Predict phase with the final explanations)? (2) Which learners still confuse mitosis with meiosis, or believe that body cells divide by meiosis to grow? Plan a short remedial session using chromosome models. (3) Did the large class limit participation, and which grouping routines (roles, two-shift gallery walk, jigsaw) worked best? (4) Which DQB questions remained unanswered, and how can they link to the next unit (for example, cancer as uncontrolled cell division, or genetics)? (5) Were Kenyan examples (ugali, sukuma wiki, mandazi, banana tissue culture in Kenya, athletes) meaningful for learners and did they help to connect the phenomenon to their lives? (6) Were safety practices with microscopes and hygiene after handling yeast/bacteria culture consistently followed? (7) How well did learners show Unity and Respect in group work and presentations?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed the baby photographs and growth time-lapse again, onion root tip cells in different stages of mitosis, chromosome models, animations of mitosis and meiosis, budding yeast, bacteria dividing by binary fission and examples of tissue culture. We also compared our Lesson 1 drawing with our final model.
What did I learn?	We learned that growth happens mainly because the number of cells increases. Existing cells copy their DNA and divide by mitosis into two identical daughter cells with the same chromosome number, so each new cell gets the right instructions. Meiosis makes gametes with half the chromosomes, and fertilisation forms a zygote that begins the growth of a new individual. Budding and binary fission are forms of cell division in yeast and bacteria, and tissue culture uses cell division to grow many plants from small pieces of tissue.
How does this explain the phenomenon?	A child grows from 3 kg to 50 kg because the zygote and later body cells divide by mitosis again and again (1, 2, 4, 8, 16...), producing billions of cells. Before each division, DNA is copied, so every daughter cell has identical chromosomes and the right

	instructions. Food such as ugali, beans and sukuma wiki supplies the raw materials and energy for making new cells, and cells then differentiate into specialised types like muscle, bone, blood and nerve cells.
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DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.